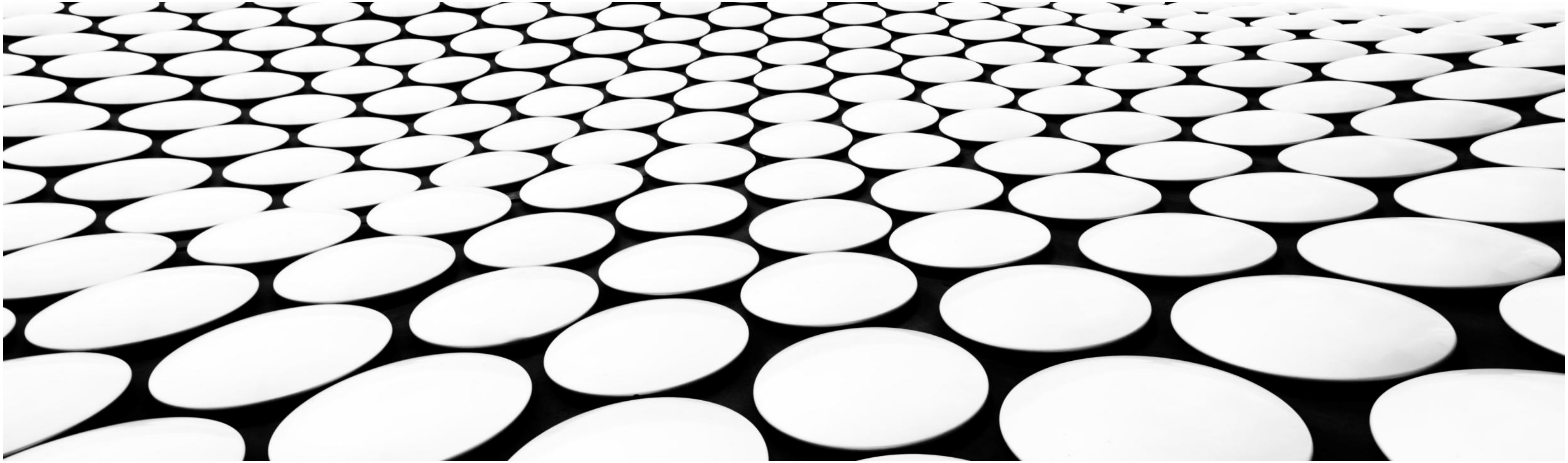
CLUSTERING

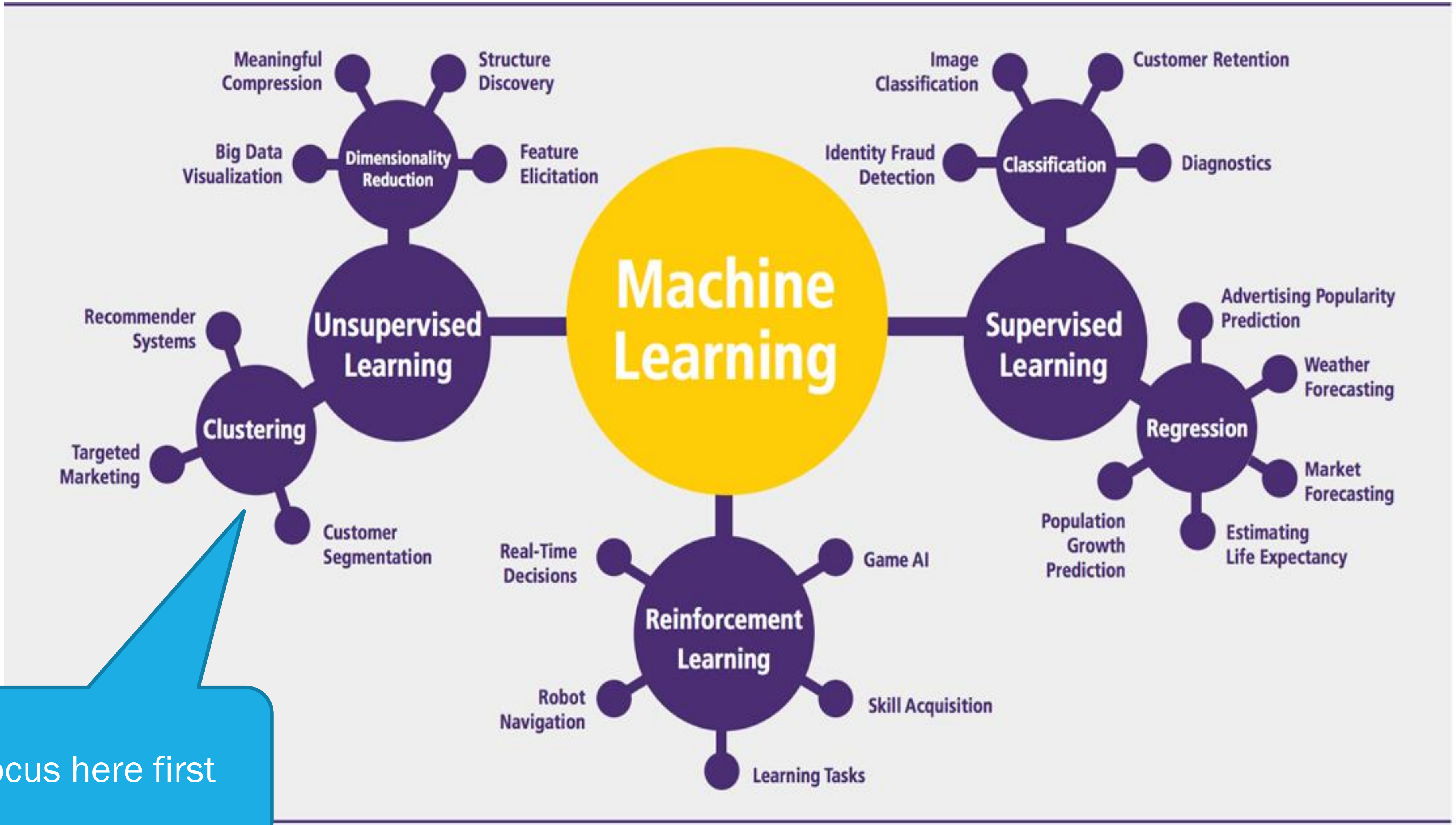
Exploring various ways of grouping data



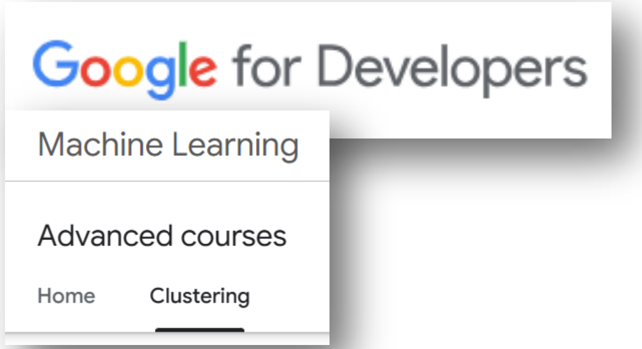
CLUSTERING

- Introduction
- Similarity measures
- Centroid-based clustering (KMeans)
- Density-based clustering (DBSCAN)

INTRODUCTION

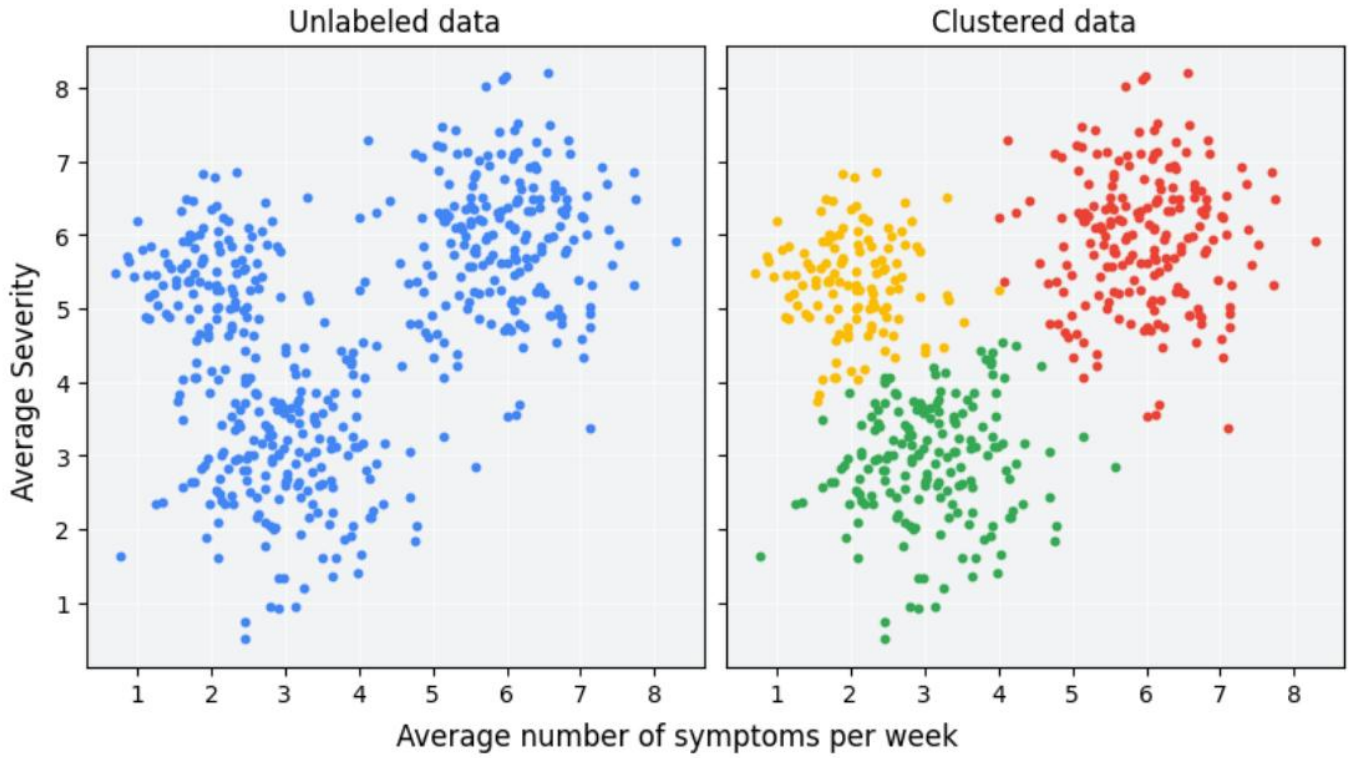


Let's focus here first



Source

Clustering is an unsupervised machine learning technique designed to group **unlabeled examples** based on their similarity to each other.



Applications

Use cases

- Patient profiling
- Market segmentation
- Music Listener profiling
- Social network analysis
- ….

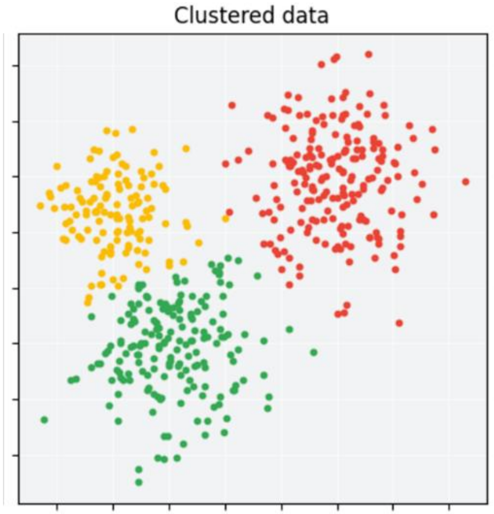
I asked ChatGPT to show me some examples with small datasets… you can do that to get more case studies

Patient Risk Profiling (Healthcare)

Patient ID	Age	BMI	Blood Pressure	Cholesterol Level
501	25	22	120/80	180
502	60	30	140/90	220
503	45	28	135/85	200
504	70	35	160/95	250
505	35	24	125/82	190
506	50	29	138/88	210
507	65	33	150/92	240

Clusters:

1. **Low-risk patients** (younger, healthy BMI, normal vitals)
2. **Moderate-risk patients** (middle-aged, slightly high BP and cholesterol)
3. **High-risk patients** (older, high BMI, severe hypertension)



Imagine in 4D space

Customer Segmentation (Marketing & Retail)

Customer ID	Age	Annual Spend ($)	Purchase Frequency
101	25	500	10
102	40	3000	50
103	30	1200	20
104	50	7000	60
105	35	2500	35
106	22		
107	45		
108	33		
109	55		
110	28		

Post-clustering observation and naming

Clusters:

1. **Low-budget shoppers** (e.g., younger customers, low spend, low frequency)
2. **Regular buyers** (e.g., middle-aged customers, moderate spend, moderate frequency)
3. **High-spending loyalists** (e.g., older customers, high spend, high frequency)

Music Recommendation (Entertainment & Media)

User ID	Genre Preference	Avg. Daily Listening Time (min)	Number of Likes
601	Pop	90	20
602	Rock	120	35
603	Jazz	60	15
604	Classical	30	5
605	Hip-Hop	100	25
606	Pop	85	18
607	Rock	130	40
608	Classical	25	

Clusters:

1. **Casual listeners** (low listening time, low engagement)
2. **Dedicated fans** (high listening time, high engagement)

Social Network Analysis (Social Media & Online Communities)

User ID	Likes Given	Comments Posted	Friends Count
301	50	10	200
302	5	2	20
303	100	30	500
304	25	5	100
305	75	20	350
306	10	1	50
307	60	15	250
308	8	0	30

Clusters:

1. **Highly engaged users**
2. **Occasional users**
3. **Inactive users**

High-level algorithm

<u>To cluster your data, you'll follow these steps:</u>

1. Prepare data.
2. Decide on a similarity metric.
3. Run clustering algorithm.
4. Interpret results and adjust your clustering.

Data cleaning / Feature Engineering

EDA – look at the data to make good choices

EDA might also help choose the clustering algorithm.

Tricky part… visual analysis, some metrics, or impact downstream

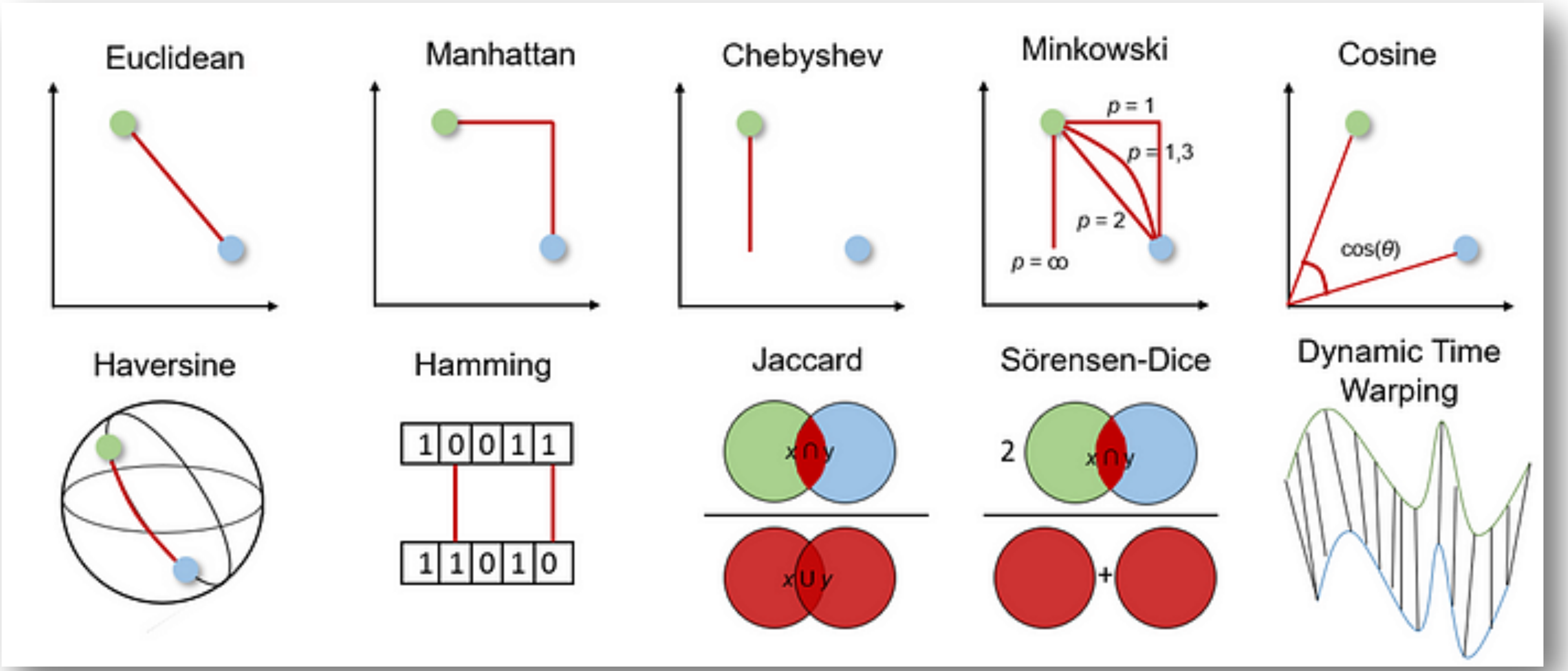
A brief introduction to Distance Measures

10 distance measures for machine learning you should have heard of

Jonte Dancker · Follow
9 min read · Oct 25, 2022

Source

Which similarity to use?



Which algorithm to use?

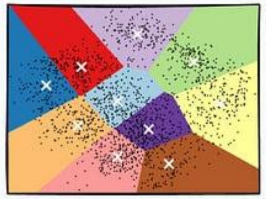
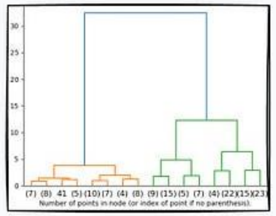
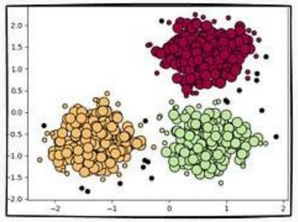
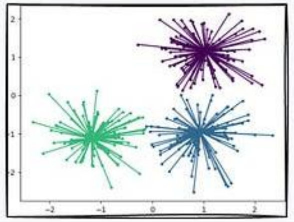
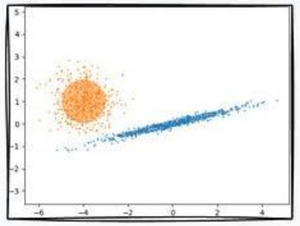
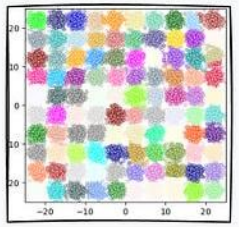
Clustering Algorithm Type		Clustering Methodology	Algorithm(s)
	Centroid-based	Cluster points based on proximity to centroid	KMeans KMeans++ KMedoids
	Connectivity-based	Cluster points based on proximity between clusters	Hierarchical Clustering (Agglomerative and Divisive)
	Density-based	Cluster points based on their density instead of proximity	DBSCAN OPTICS HDBSCAN

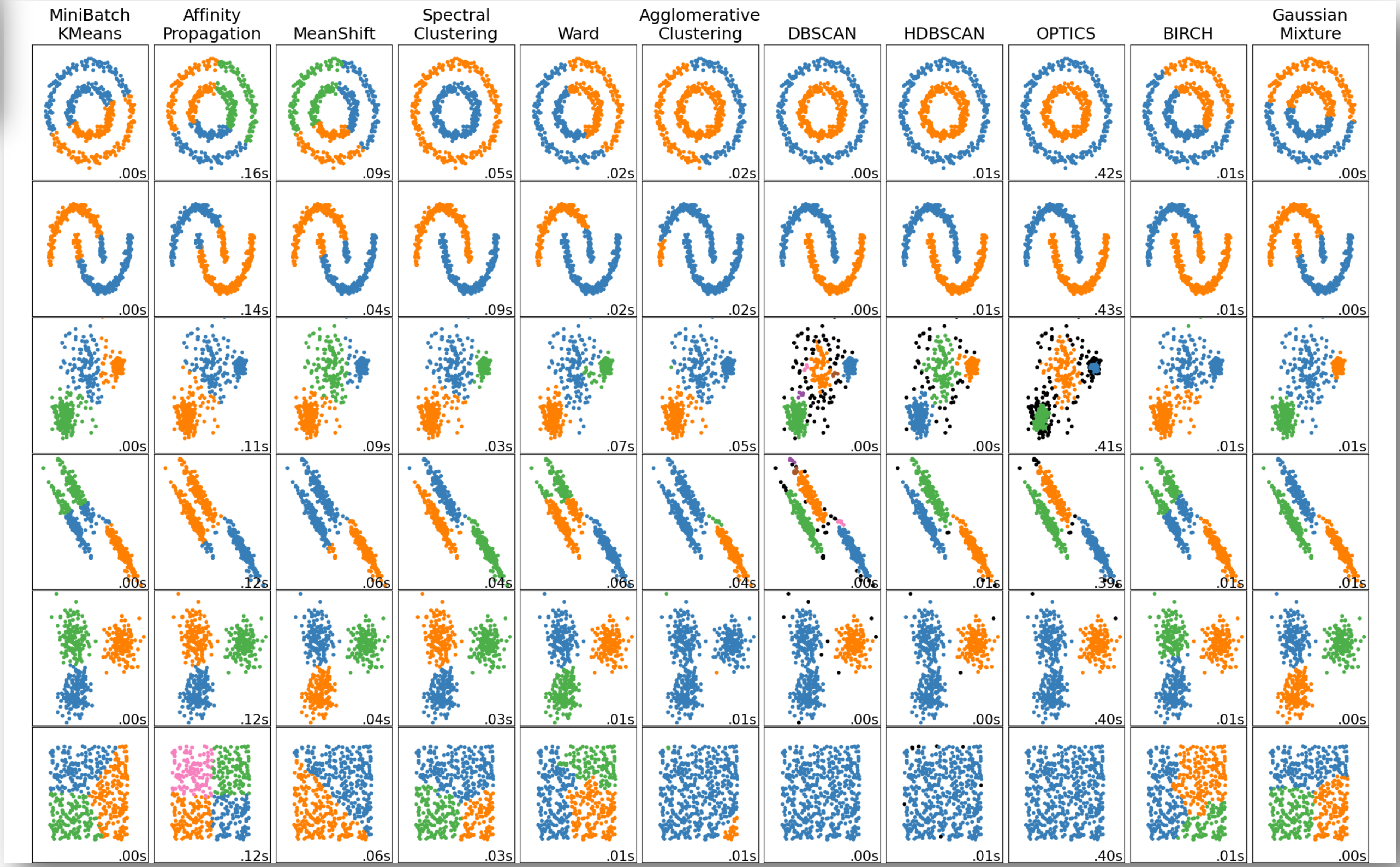
Image Source (not full paid article)

Which algorithm to use?

Clustering Algorithm Type		Clustering Methodology	Algorithm(s)
	Graph-based	Cluster points based on graph distance	Affinitiy Propagation Spectral Clustering
	Distribution-based	Cluster points based on their likelihood of belonging to the same distribution.	Gaussian Mixture Models (GMMs)
	Compression-based	Transform data to a lower dimensional space and then perform clustering	BIRCH

[Source]

Different algorithms on different datasets – Nicely done



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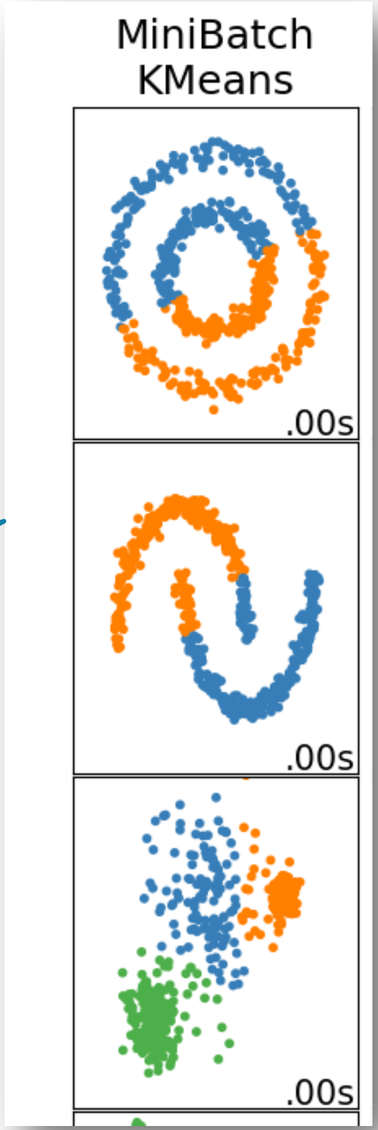
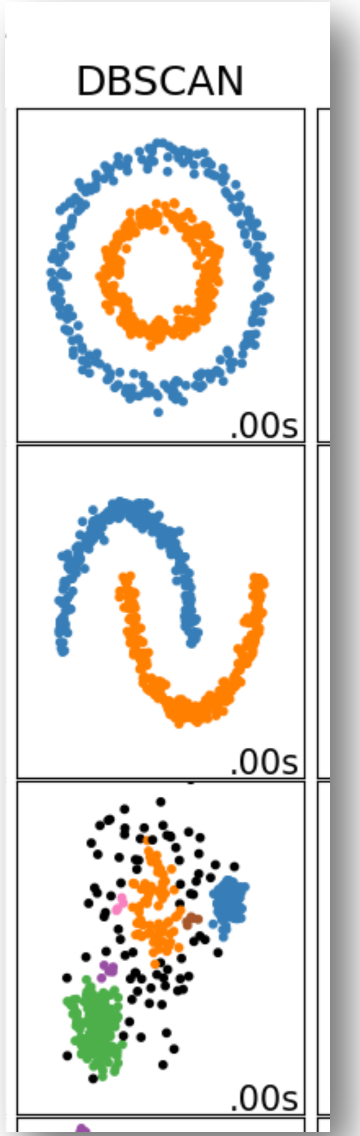




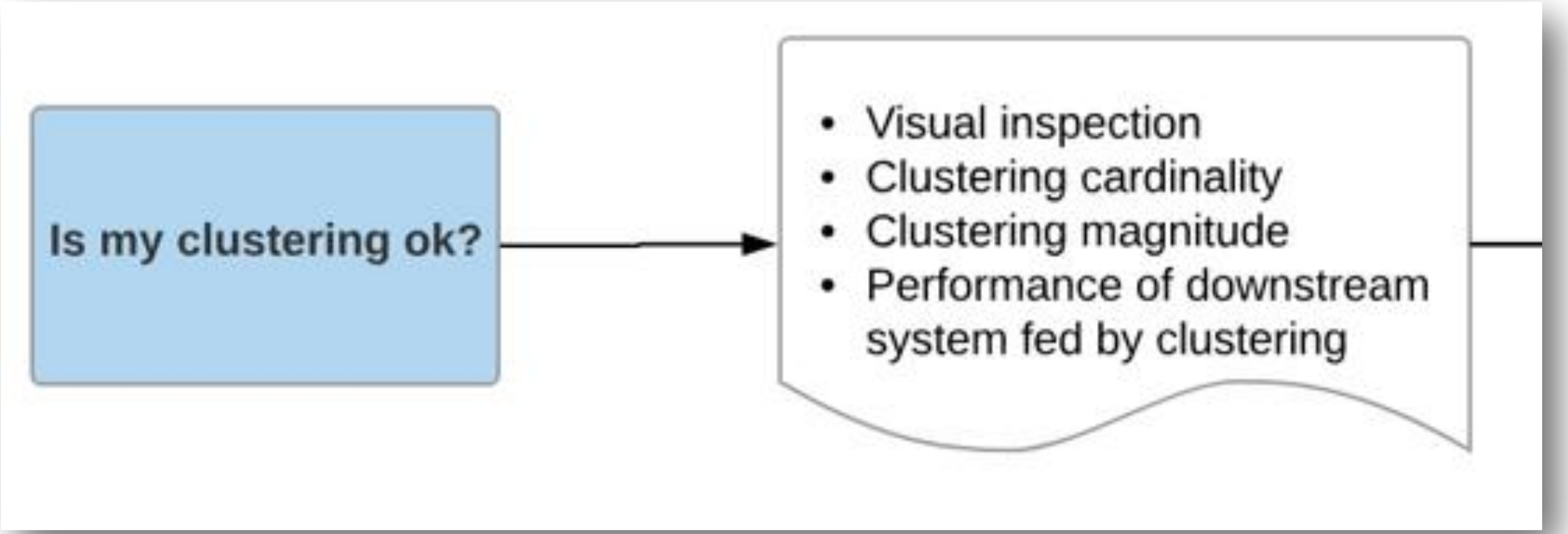
Density-based
DBSCAN

Centroid-based
KMeans

To cluster your data, you'll follow these steps:

1. Prepare data.
2. Decide on a similarity metric.
3. Run clustering algorithm.
4. Interpret results and adjust your clustering.

How to evaluate?



Similarity measures

A brief introduction to Distance Measures

10 distance measures for machine learning you should have heard of

Jonte Dancker · Follow
9 min read · Oct 25, 2022

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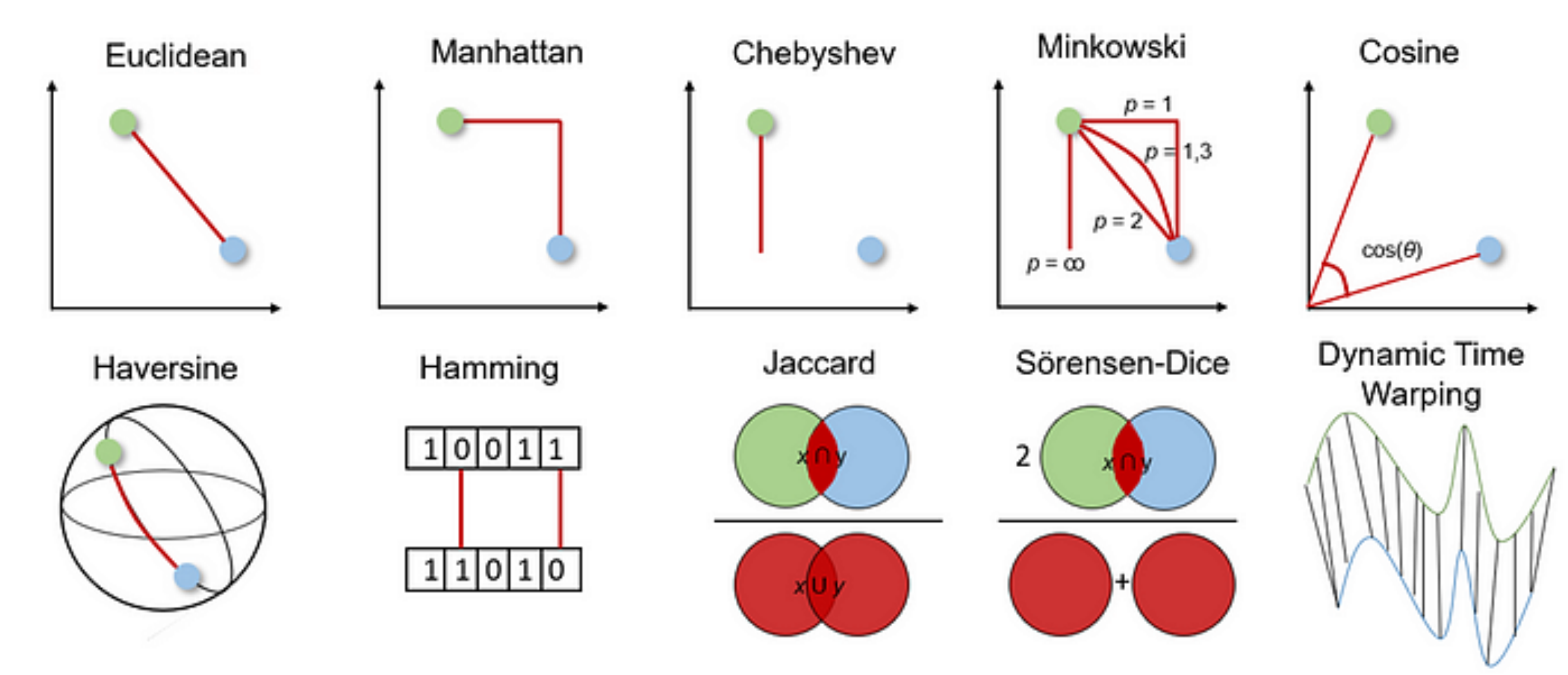
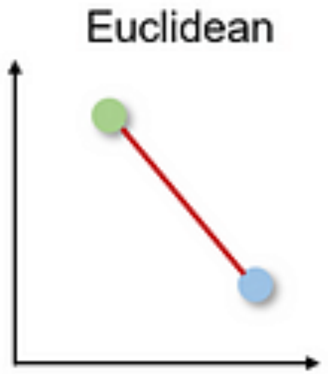
Again I've asked ChatGPT for some examples... you can continue



Euclidian

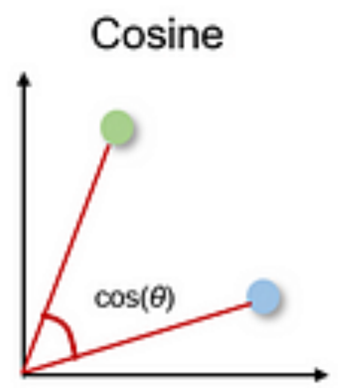
$$d = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$$



Student ID	Math Score	Science Score	English Score
201	85	78	88
202	60	65	55
203	92	90	95
204	75	80	70
205	50	55	48
206	88	85	90

Cosine

$$\text{Similarity} = \frac{A \cdot B}{||A|| \cdot ||B||}$$



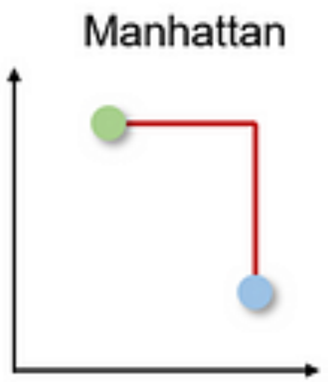
User ID	Movie A	Movie B	Movie C	Movie D	Movie E	Movie F
User 1	5	4	0	3	2	0
User 2	3	2.5	0	1.5	1	0
User 3	4	0	5	2	0	3
User 4	0	5	4	3	0	2

not interested in magnitude of vectors

Manhattan

$$d_{Manhattan} = |x_1 - x_2| + |y_1 - y_2|$$

absolute distances

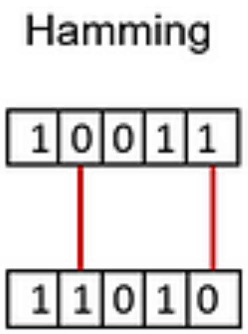


Location	X (Blocks East)	Y (Blocks North)
A	3	8
B	7	2
C	5	6
D	8	3

Now, assume our **warehouse is at (2,2)**.

Hamming

Edit distance inspired by Hamming distance



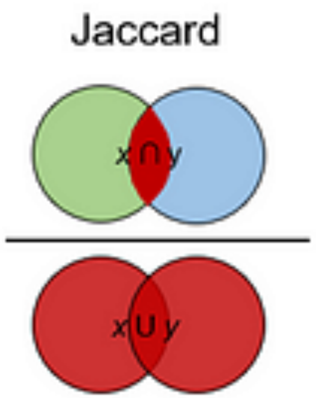
The **Hamming distance** between two strings is the number of positions at which the characters differ. We will compare each position in the two sequences:

Sequence	DNA Sequence
A	ATCGGTCGTA
B	ATCAGTCGGA
C	ATCGGTCGAA
D	ATCAGTCGTA
E	GTAGGTCGTA

2

Jaccard

$$\text{Jaccard Distance} = 1 - \frac{|A \cap B|}{|A \cup B|}$$



Dataset: Movie Genres Liked by Users

User	Movie Genres
User 1	Action, Comedy, Drama
User 2	Action, Romance, Drama
User 3	Comedy, Romance, Thriller
User 4	Action, Comedy

Centroid-based clustering
KMeans

The **centroid** of a cluster is the arithmetic mean of all the points in the cluster.

Centroid-based clustering organizes the data into non-hierarchical clusters.

Centroid-based clustering algorithms are efficient but sensitive to initial conditions and outliers. Of these, k-means is the most widely used. It requires users to define the number of centroids, *k*, and works well with clusters of roughly equal size.

Additional resources, specifically on KMeans

Performing K-means Clustering with Python and Scikit-learn

Francesco Franco · Follow
Published in Artificial Intelligence in Plain English · 11 min read · Dec 27, 2024

Source

KMeans Clustering: A Step-by-Step Guide

Debmalya Mondal · Follow
5 min read · Nov 5, 2024

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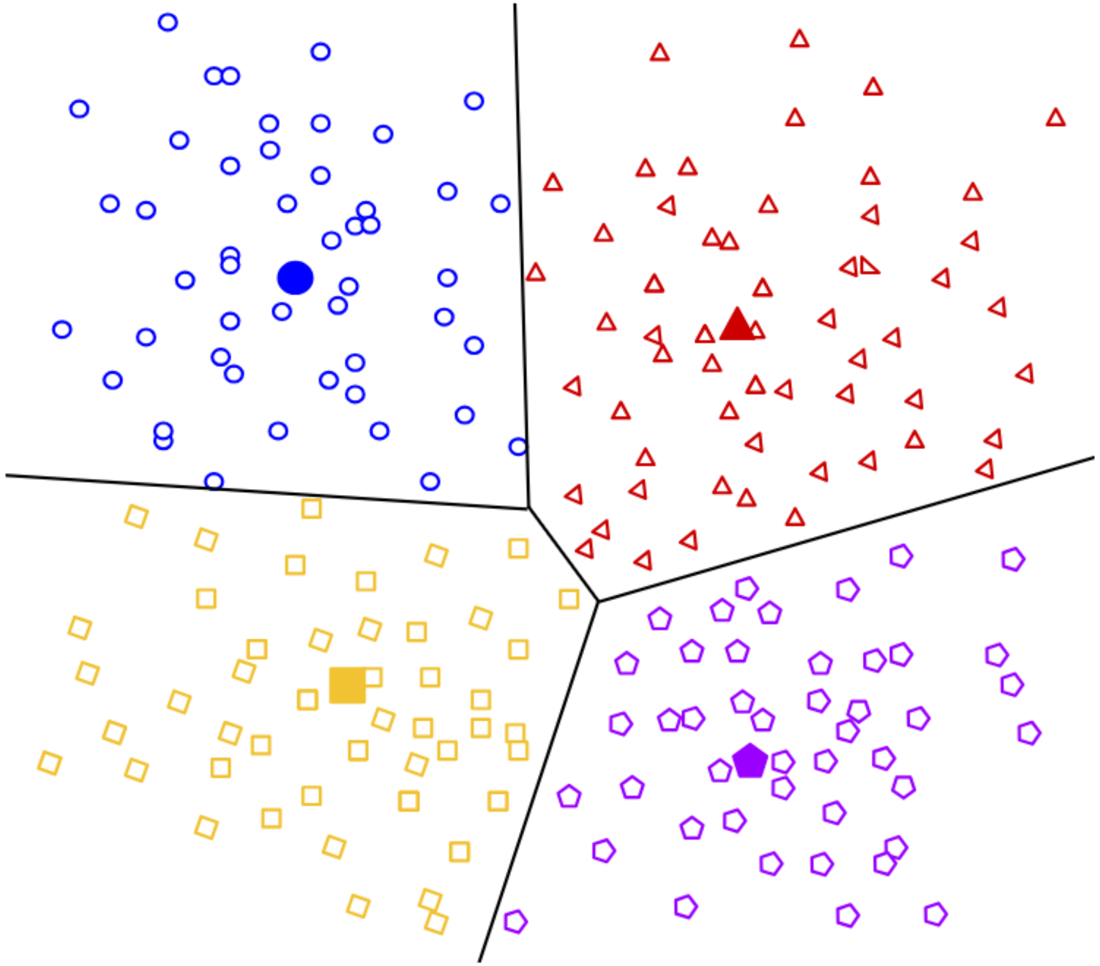


Figure 1: Example of centroid-based clustering.

Example of a result with 4 clusters

Kmeans Algorithm

Initialization
1. Provide an initial guess for k, which can be revised later.
2. Randomly choose k centroids.
3. Assign each point to the nearest centroid to get k initial clusters.

Repeat
    4. For each cluster, calculate a new centroid by taking the mean position of all points in the cluster.
    5. Reassign each point to the cluster having the nearest new centroid.

Until points no longer change clusters.

In the case of large datasets, you can stop the algorithm before convergence based on other criteria

1. Provide an initial guess for k, which can be revised later.
2. Randomly choose k centroids

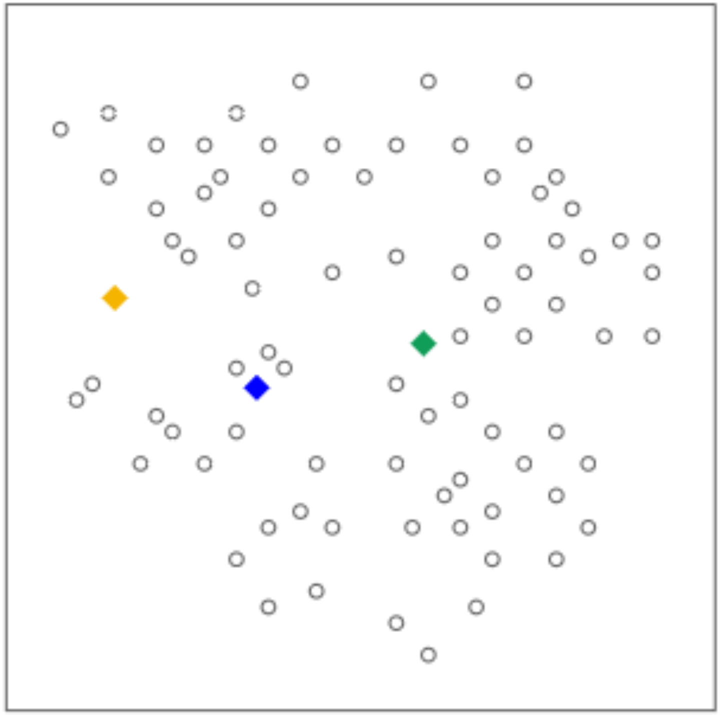
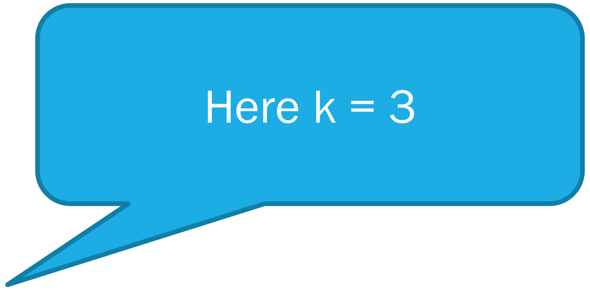


Figure 1: k-means at initialization.



3. Assign each point to the nearest centroid to get k initial clusters.

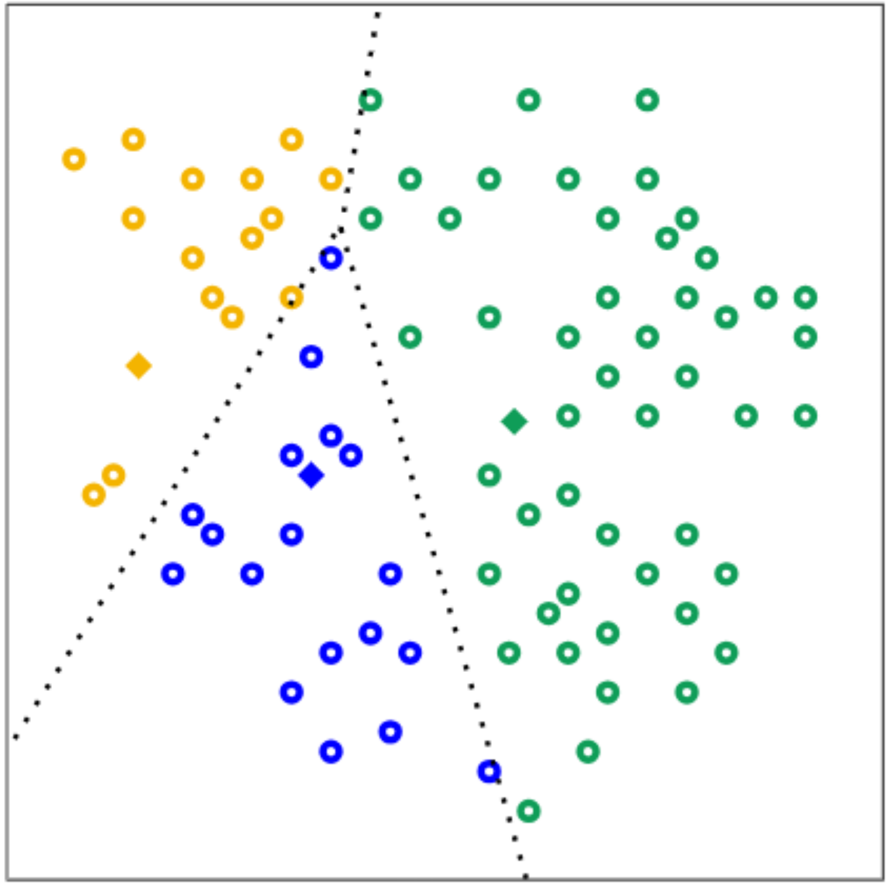


Figure 2: Initial clusters.

4. For each cluster, calculate a new centroid by taking the mean position of all points in the cluster.

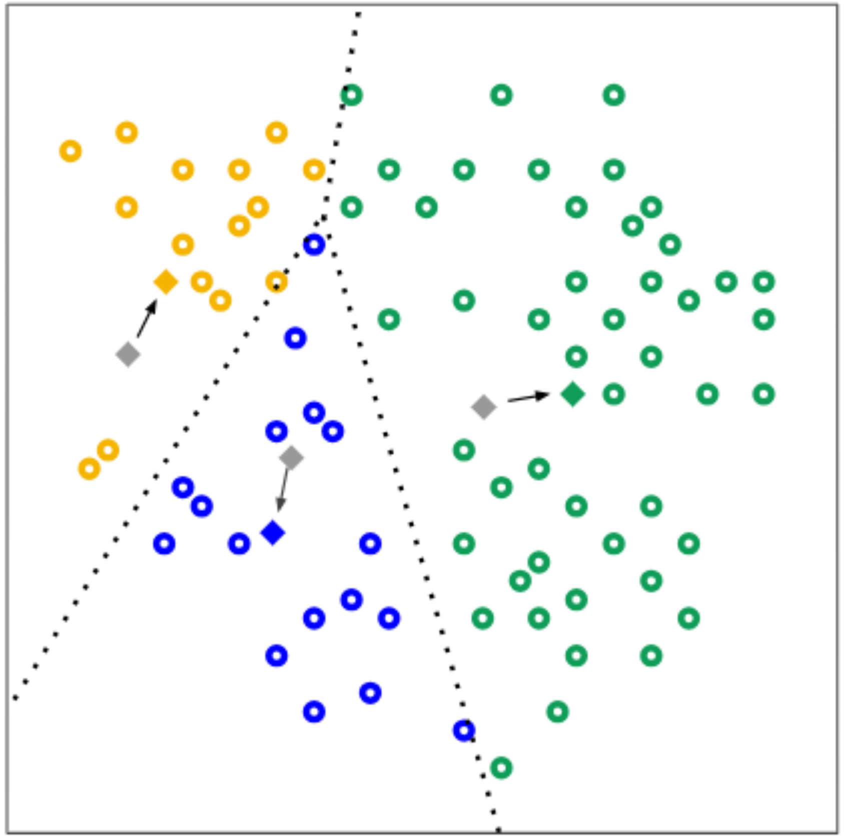


Figure 3: Recomputed centroids.

5. Reassign each point to the nearest new centroid.

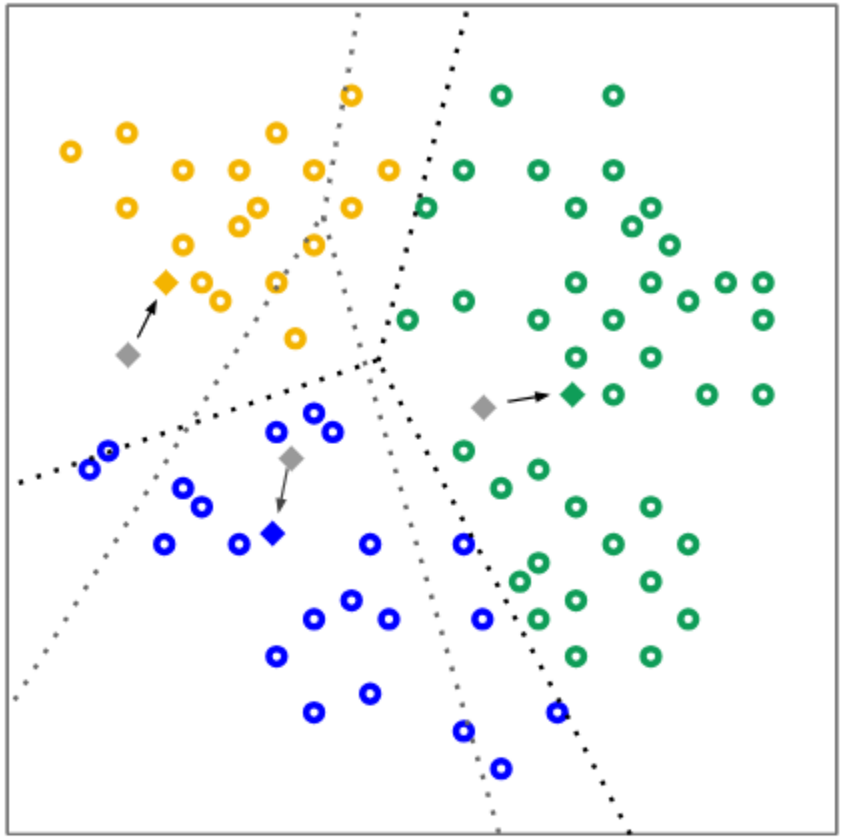


Figure 4: Clusters after reassignment.

6. Repeat steps 4 and 5, recalculating centroids and cluster membership, until points no longer change clusters.

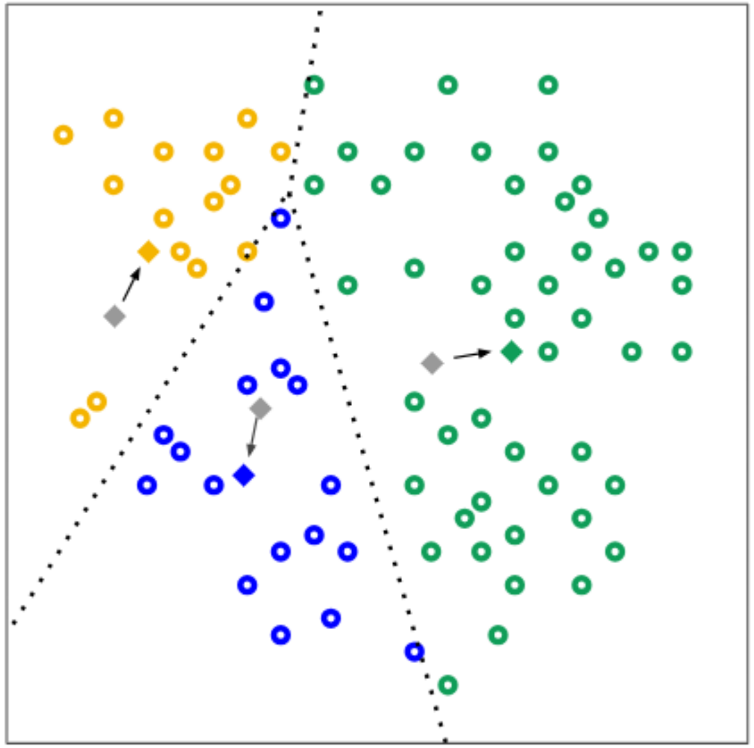


Figure 3: Recomputed centroids.

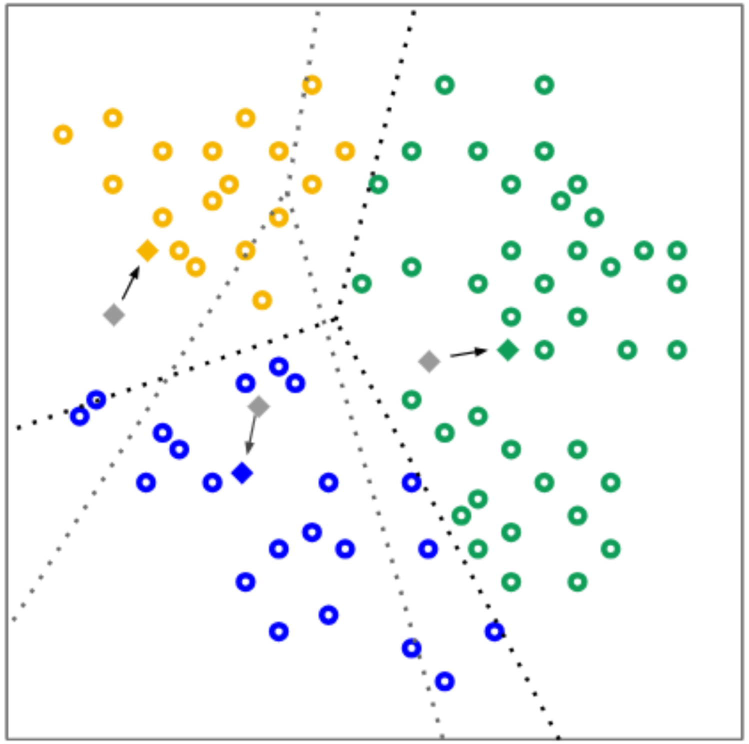


Figure 4: Clusters after reassignment.

Because the centroid positions are initially chosen at random, k-means can return significantly different results on successive runs.

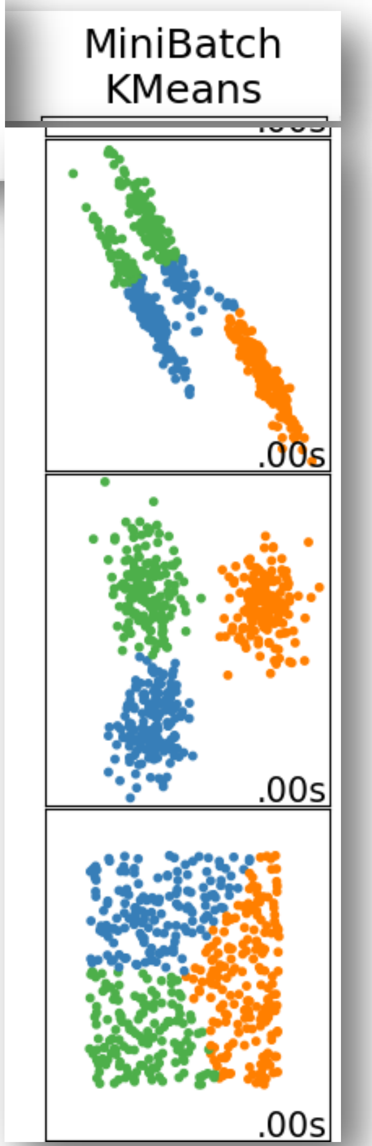
To solve this problem, run k-means multiple times and choose the result with the best quality metrics.

The variation mini-batch KMeans might help convergence and stability

ML | Mini Batch K-means clustering algorithm

Last Updated : 02 Jun, 2024

Source



Mini Batch K-means algorithm‘s main idea is to use small random batches of data of a fixed size, so they can be stored in memory.

Each iteration a new random sample from the dataset is obtained and used to update the clusters and this is repeated until convergence.

Evaluation

Because clustering is unsupervised, no ground truth is available to verify results. The absence of truth complicates assessments of quality. Moreover, real-world datasets typically don't offer obvious clusters of examples as nicely defined as in the example below.

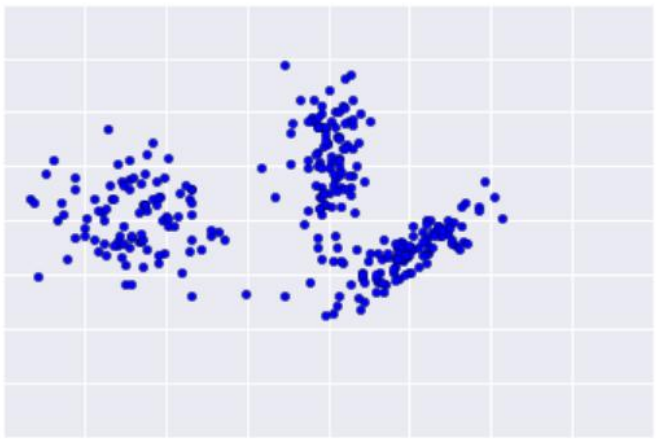


Figure 1: An ideal data plot. Real-world data rarely looks like this.

Visual inspection

Check that the clusters look as you'd expect, and that examples you consider similar to each other appear in the same cluster.

Cluster Metrics

Possibly use these commonly-used metrics (not an exhaustive list):
- Cluster cardinality
- Cluster magnitude
- Overall Distance

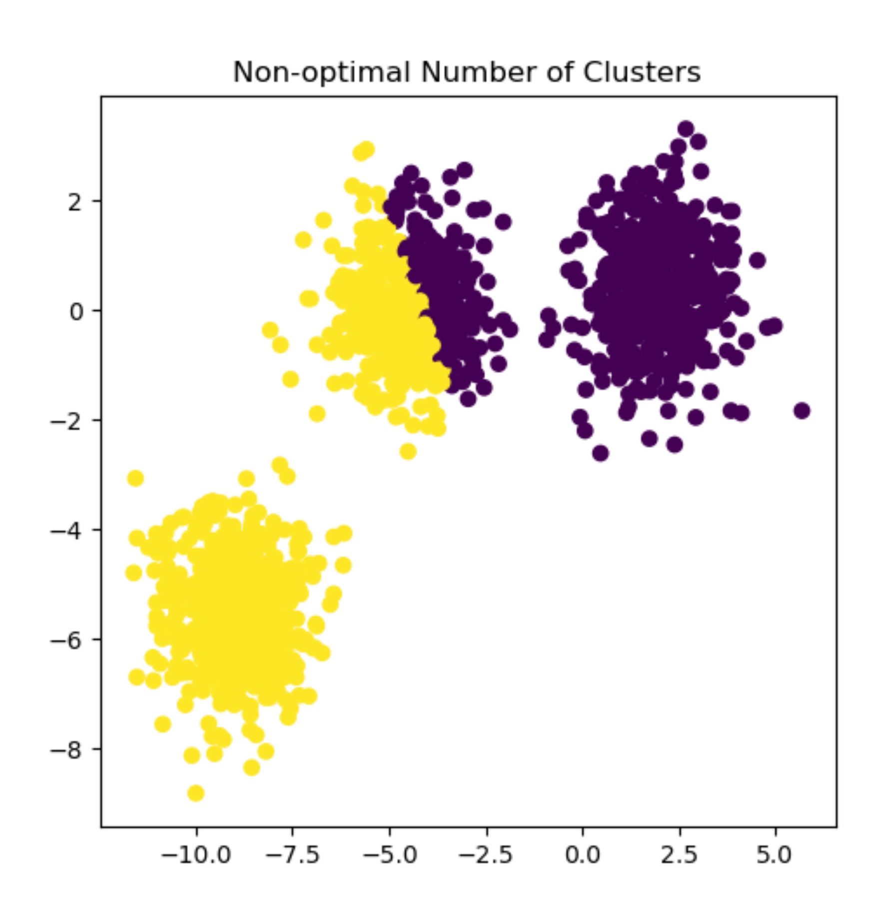
Similarity Measure

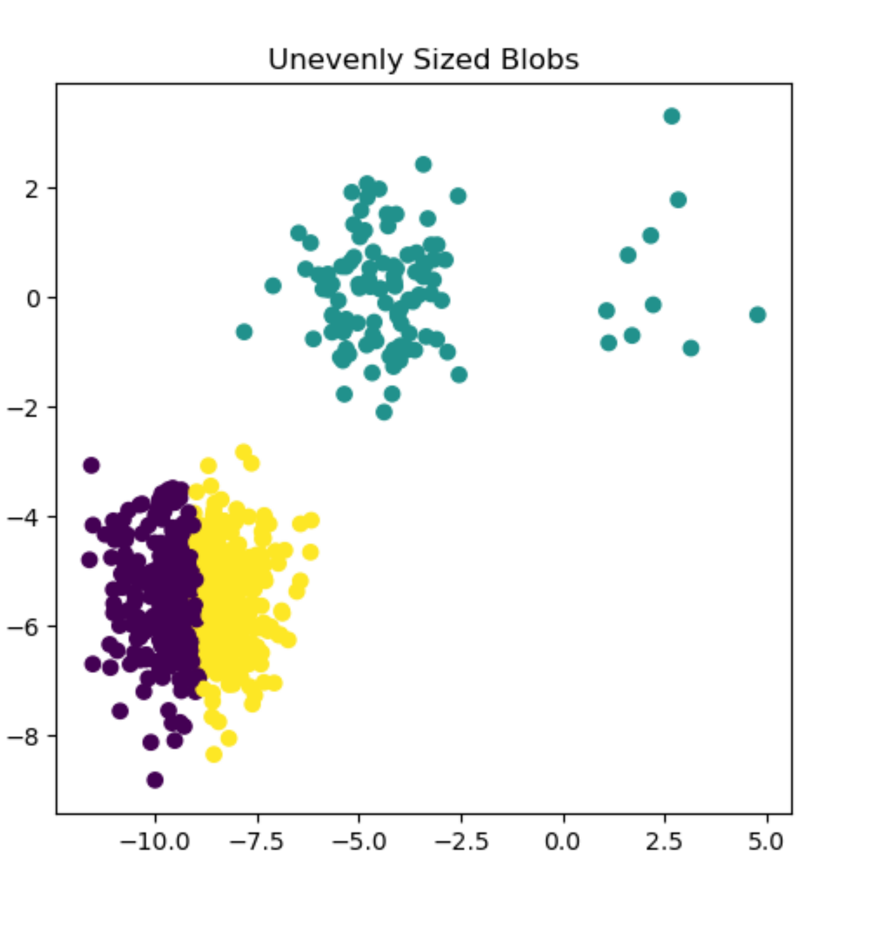
If the clustering seems inadequate, perhaps reinvestigate the similarity measure.

Impact

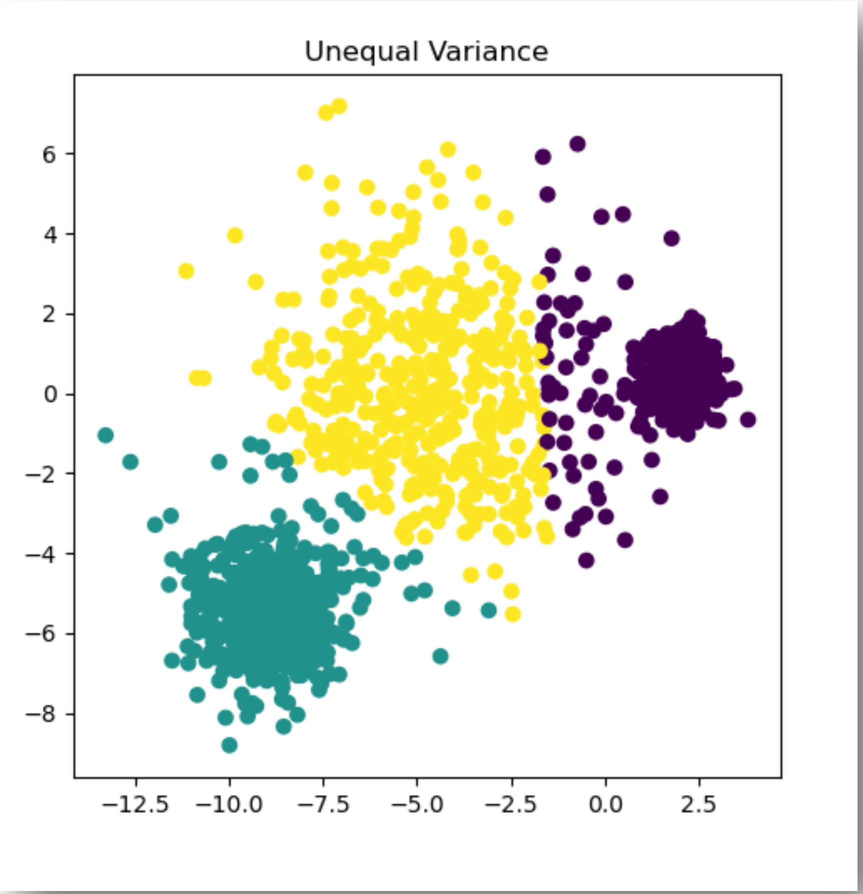
Evaluate the downstream performance.

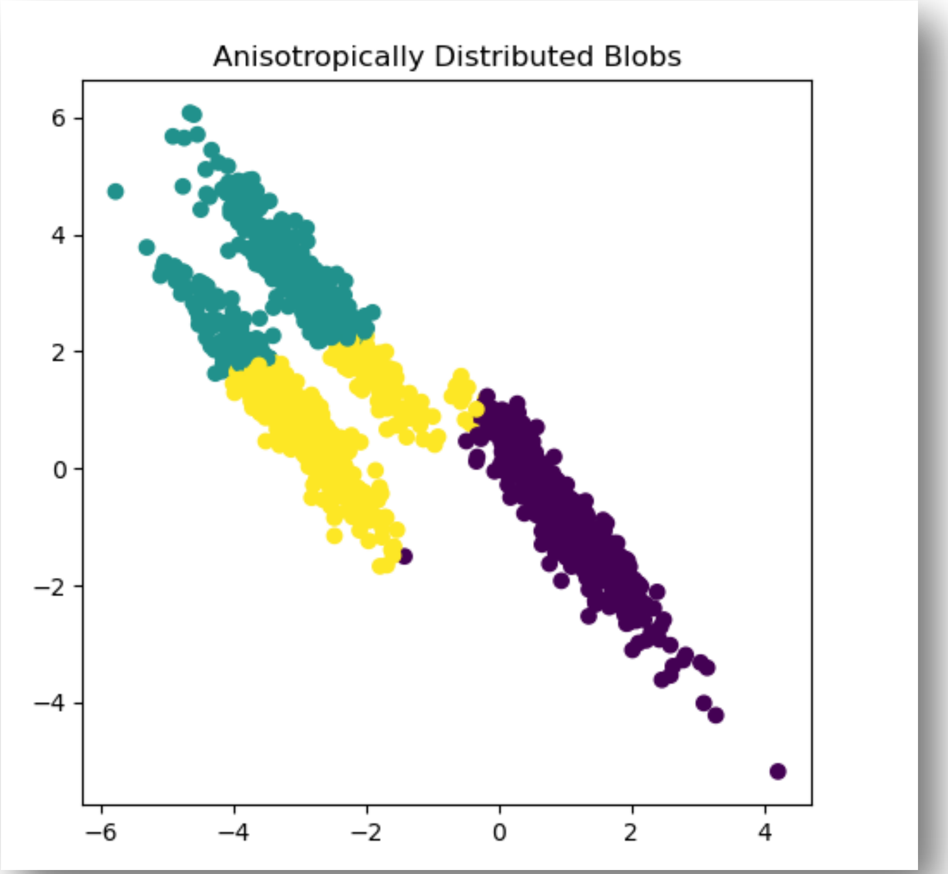
Visual Inspection





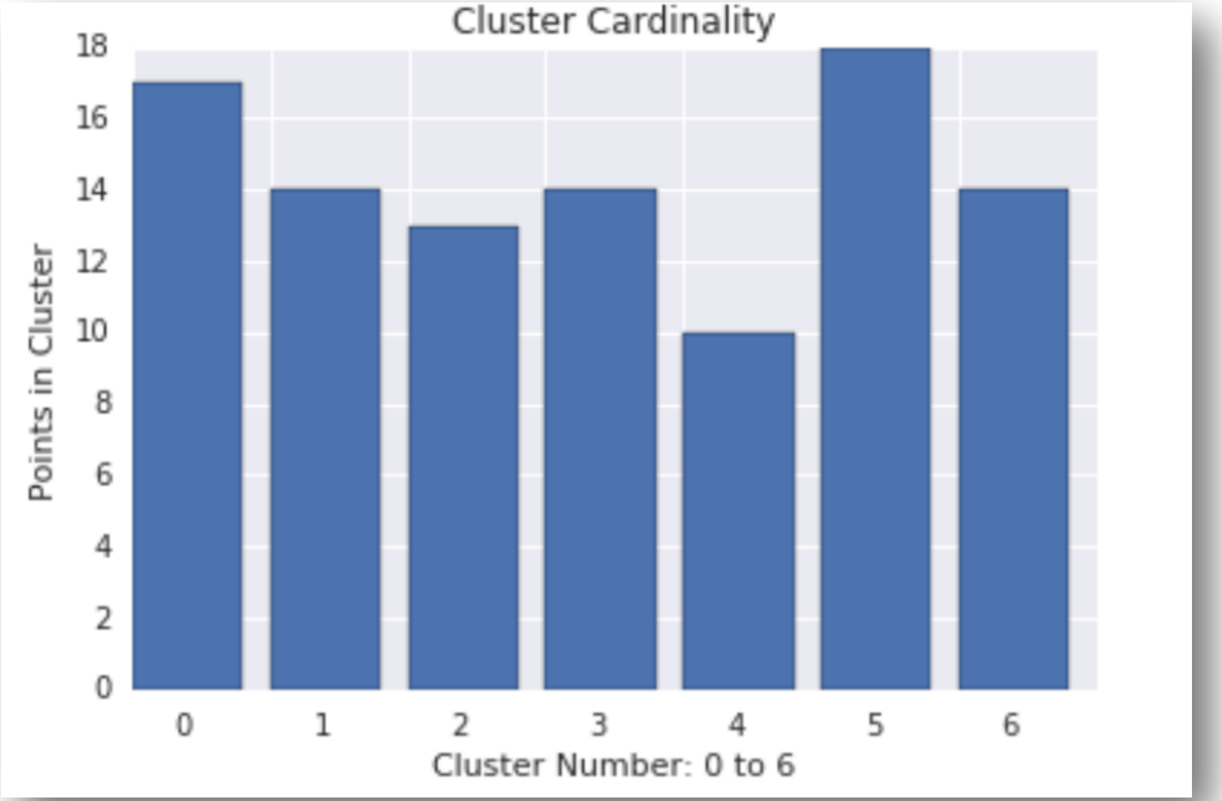
Visual Inspection





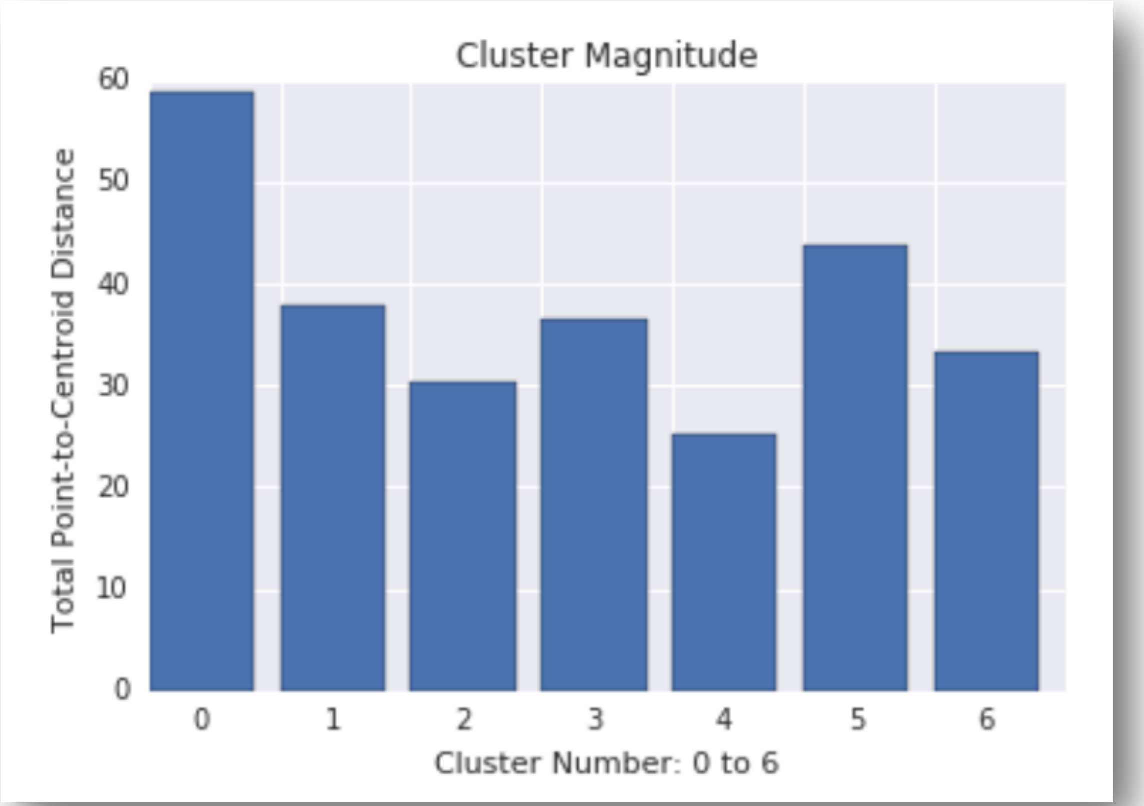
Cluster metrics

Cluster cardinality is the number of examples per cluster. Plot the cluster cardinality for all clusters and investigate clusters that are major outliers.



Cluster metrics

Cluster magnitude is the sum of distances from all examples in a cluster to the cluster's centroid. Plot cluster magnitude for all clusters and investigate outliers.



Overall Distance : Find the optimal number of clusters

Cluster Metrics

Try running the algorithm with increasing values of k and note the sum of all cluster magnitudes.

As k increases, clusters become smaller, and the total distance of points from centroids decreases. We can treat this total distance as a loss. Plot this distance against the number of clusters.

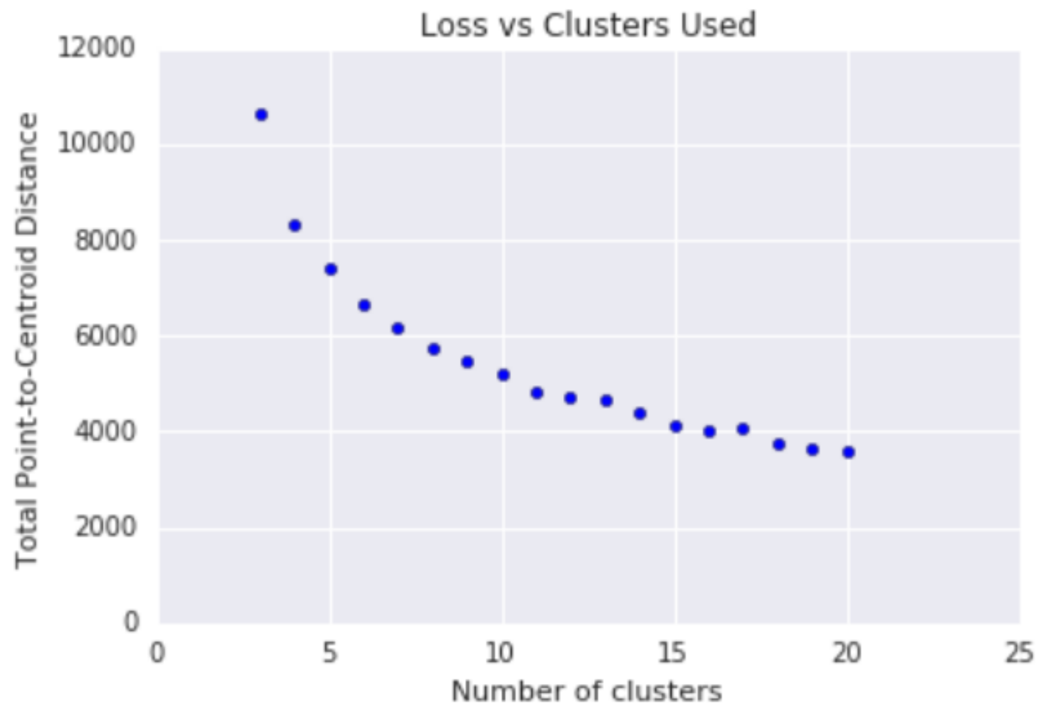
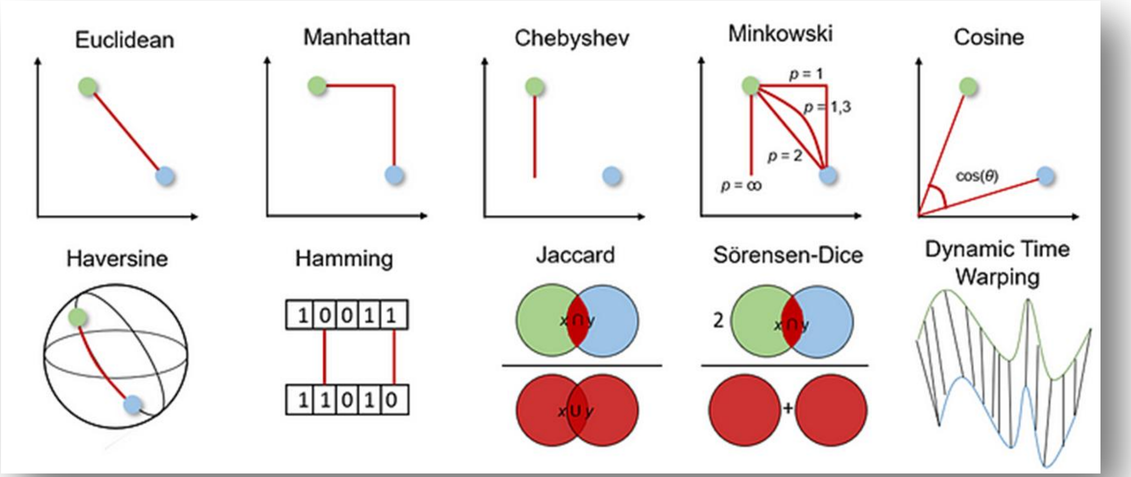


Figure 5: Loss versus number of clusters

Similarity Measure

Your clustering algorithm is only as good as your similarity measure. Make sure your similarity measure returns sensible results.

A quick check is to identify pairs of examples known to be more or less similar. Calculate the similarity measure for each pair of examples and compare your results to your knowledge: pairs of similar examples should have a higher similarity measure than pairs of dissimilar examples.



Impact

3. Music Recommendation (Entertainment & Media)

User ID	Genre Preference	Avg. Daily Listening Time (min)	Number of Likes
601	Pop	90	20
602	Rock	120	35
603	Jazz	60	15
604	Classical	30	5
605	Hip-Hop	100	25
606	Pop	85	18
607	Rock	130	40
608	Classical	25	

Would the clustering help the Recommendation system?

Clusters:

1. **Casual listeners** (low listening time, low engagement)
2. **Dedicated fans** (high listening time, high engagement)

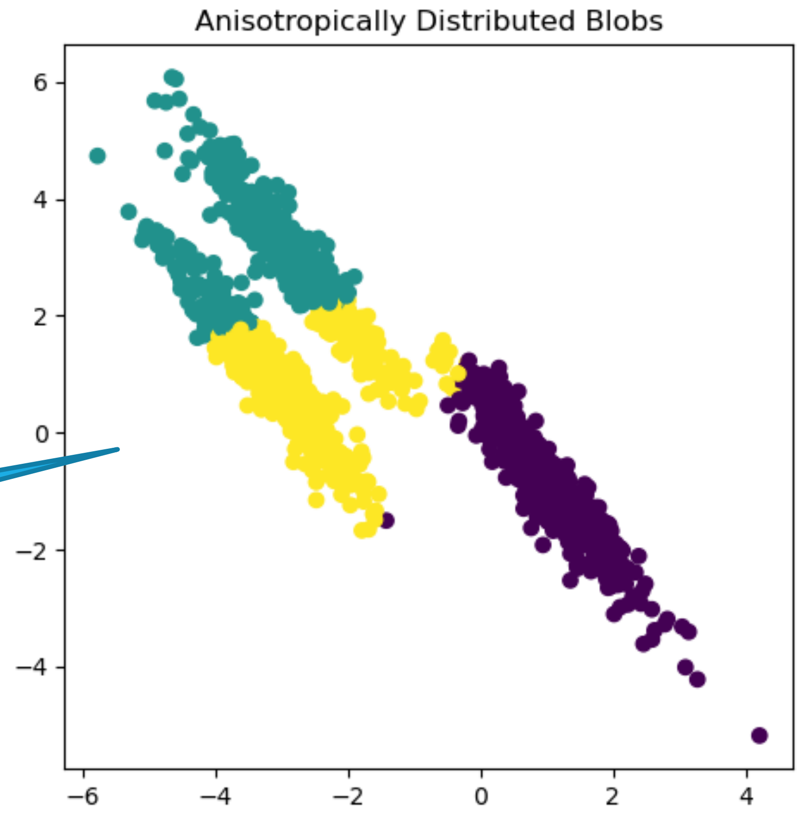
Density-based clustering *DBSCAN*

Density-based clustering connects contiguous areas of high example density into clusters. This allows for the discovery of any number of clusters of any shape.

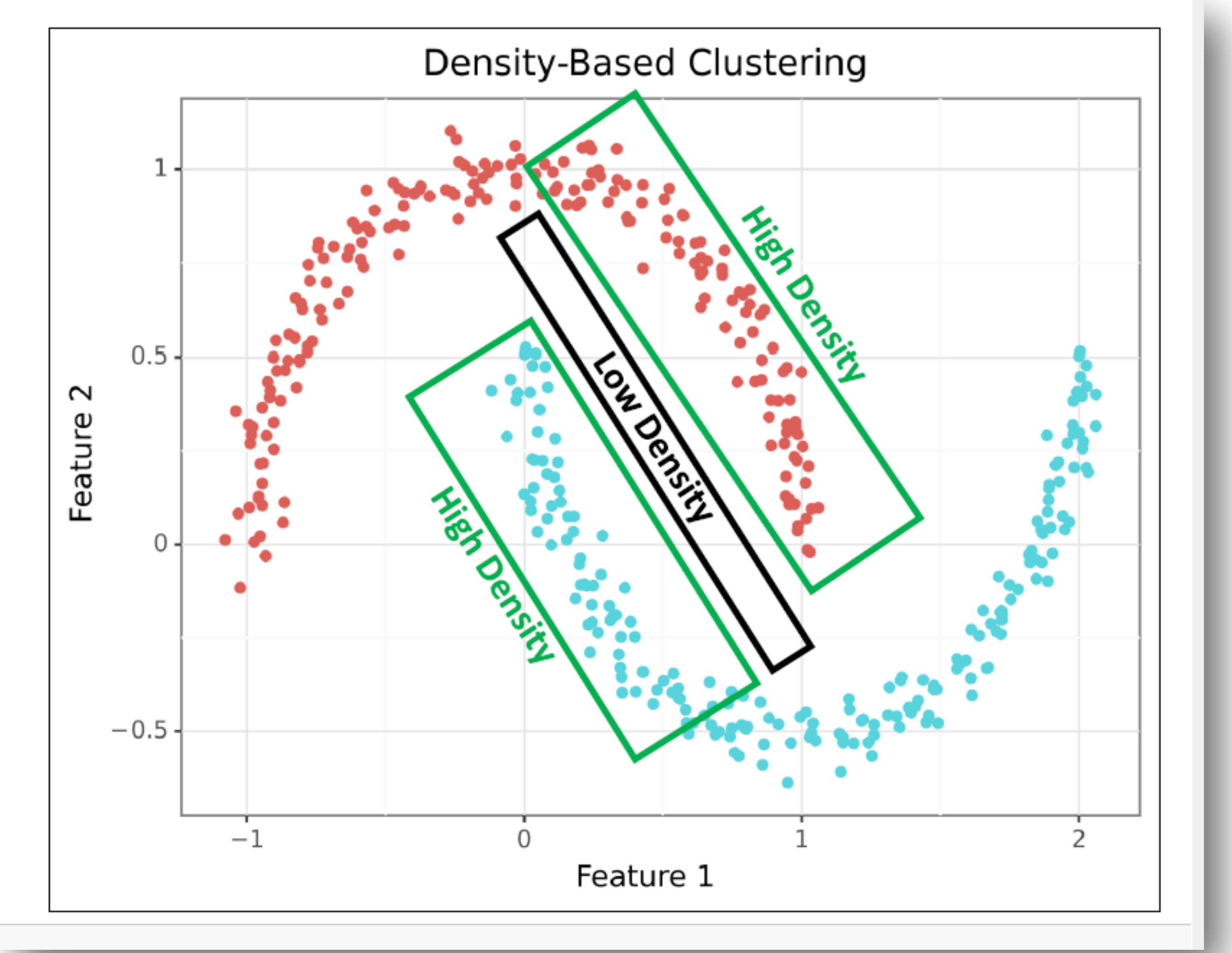
Do not specify outliers ahead of time

Outliers are not assigned to clusters. These algorithms have difficulty with clusters of different density and data with high dimensions.

Can find outliers

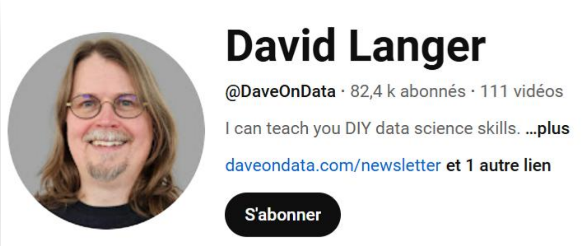


DBSCAN will be more appropriate for this type of data distribution



Density-based clusters are regions dense with data points surrounded by regions with a low density of data points.

Density-based clusters are used when clusters are irregular (e.g., not spherical), intertwined, and/or when outliers are present.

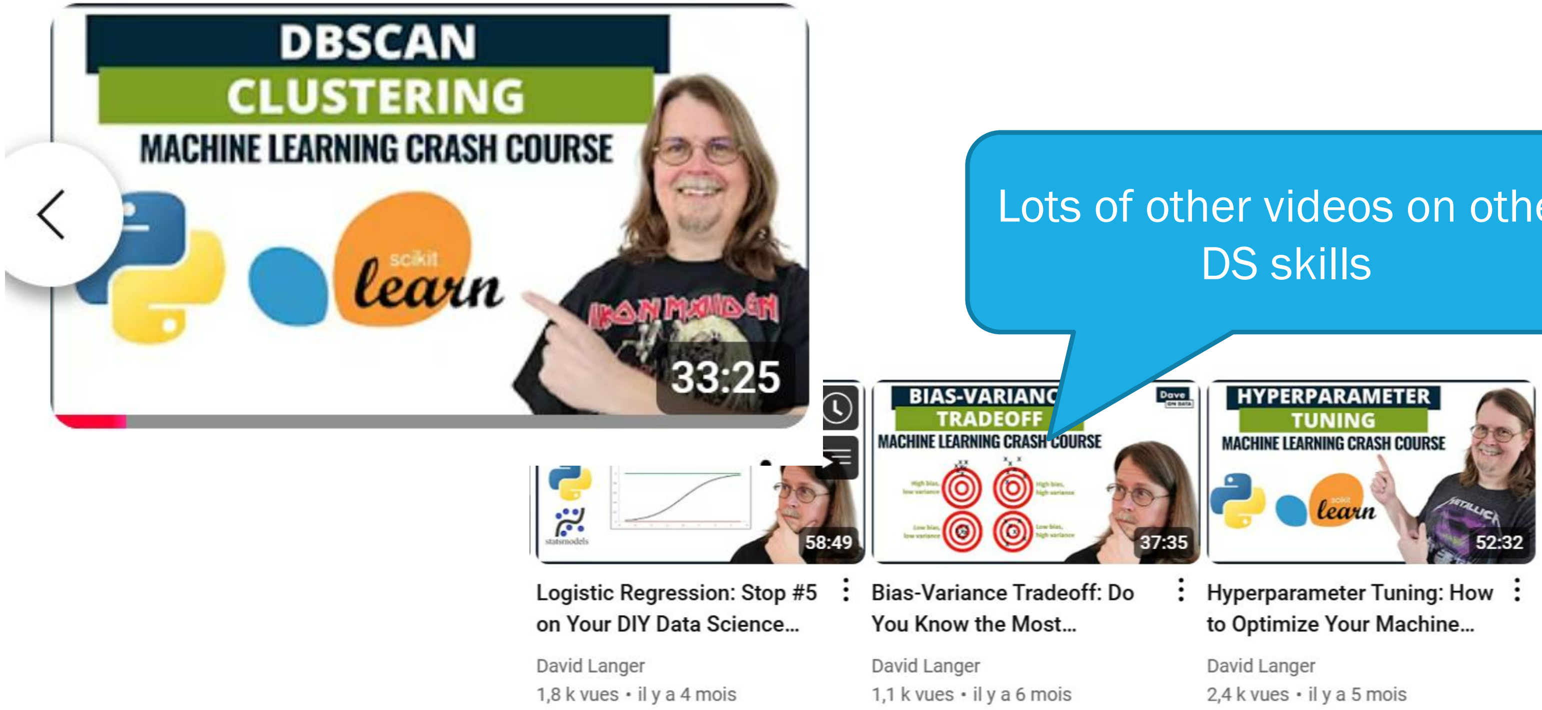


https://www.youtube.com/@DaveOnData

Clustering Like a Pro: A Beginner's Guide to DBSCAN

Sachinsoni · Follow
7 min read · Dec 26, 2023

Source

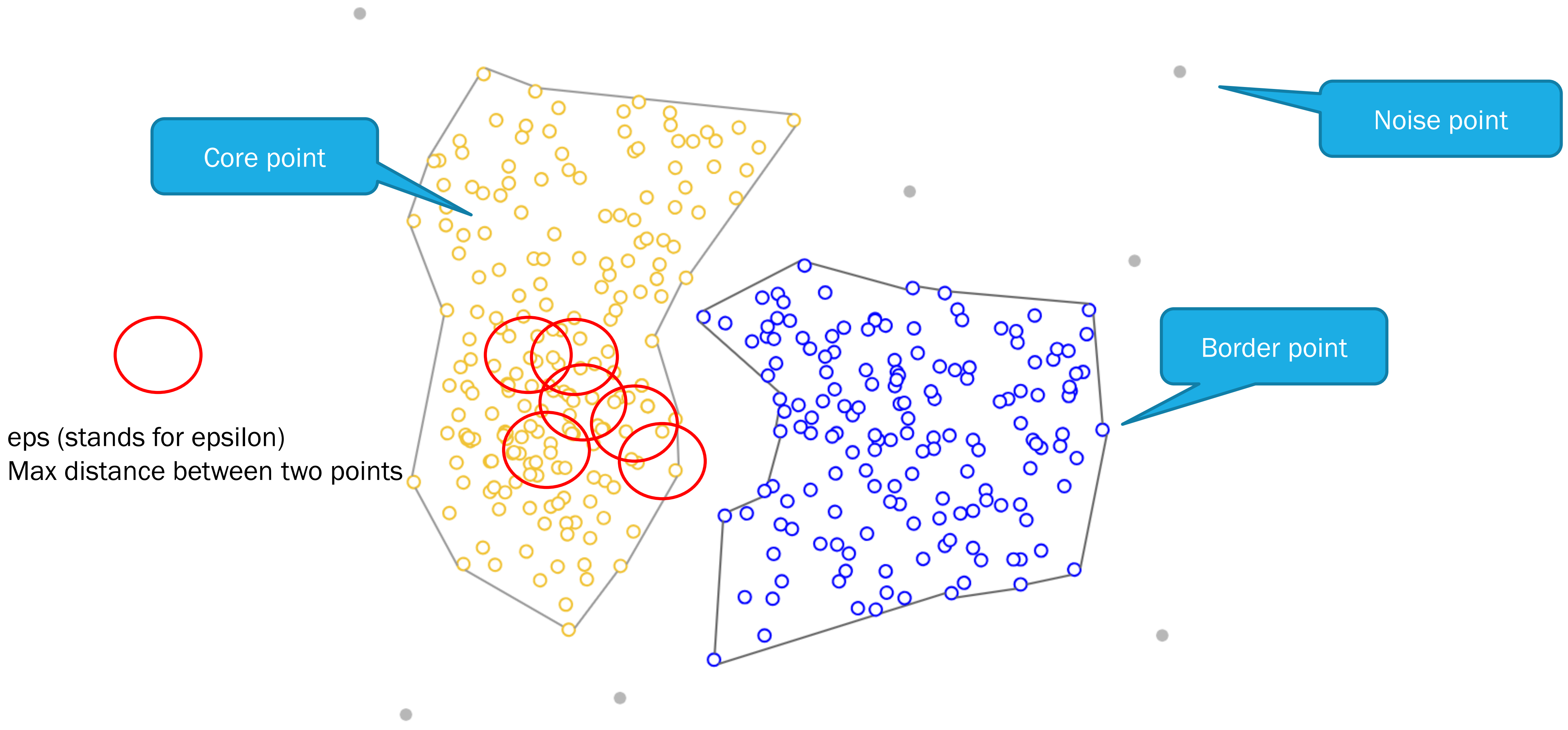


Lots of other videos on other DS skills

Terminology

The DBSCAN algorithm is built according to the following definitions:

- **Core points**: These are data points inside a cluster. A data point is core if it is surrounded by a minimum number of other data points within a set distance. When you use DBSCAN you set the distance (known as **eps**) and the minimum number of points within the distance (known as **min_samples**).
- **Border points**: A border point is not a core point, but falls within the set distance (i.e., eps) of a core point.
- **Noise points**: A noise point is any point that is neither a core point nor a border point.

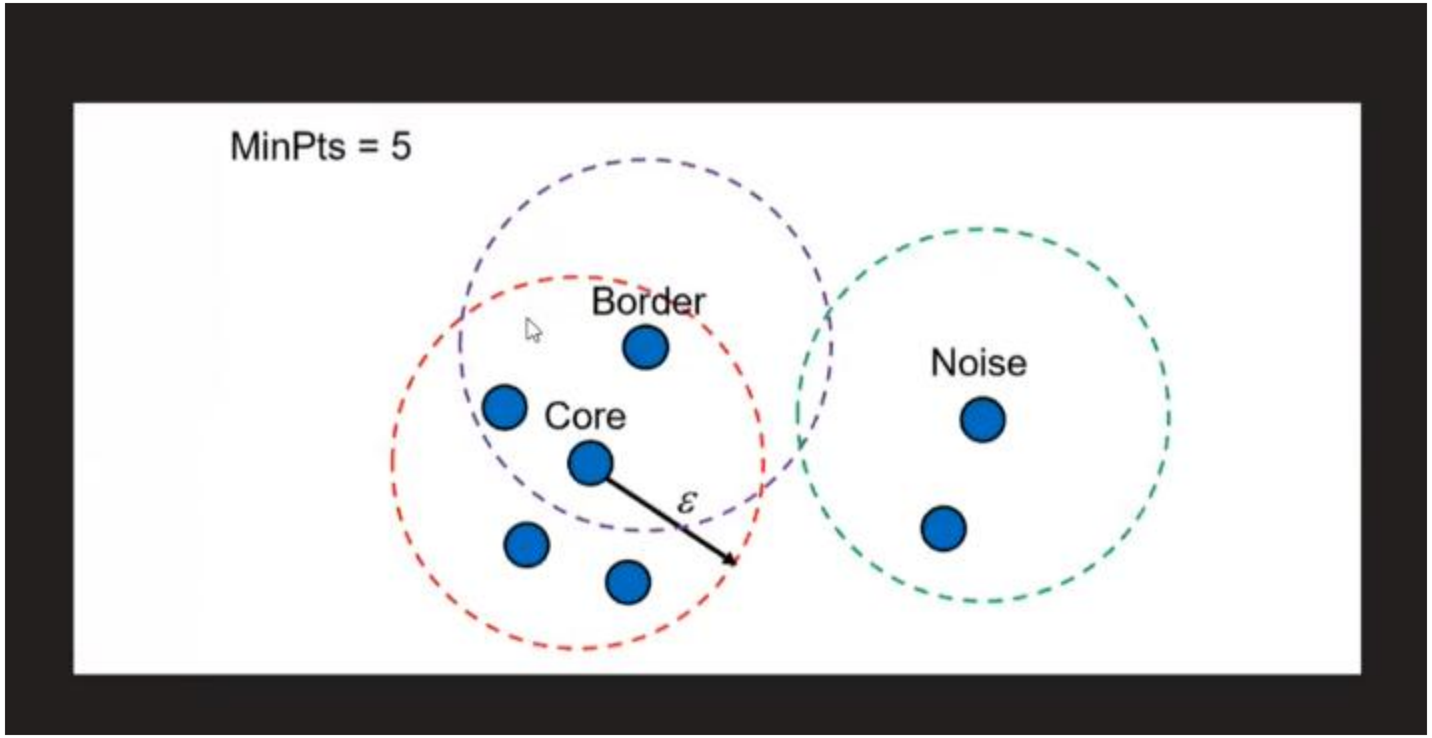

Core point
Noise point
Border point
eps (stands for epsilon)
Max distance between two points

Algorithm

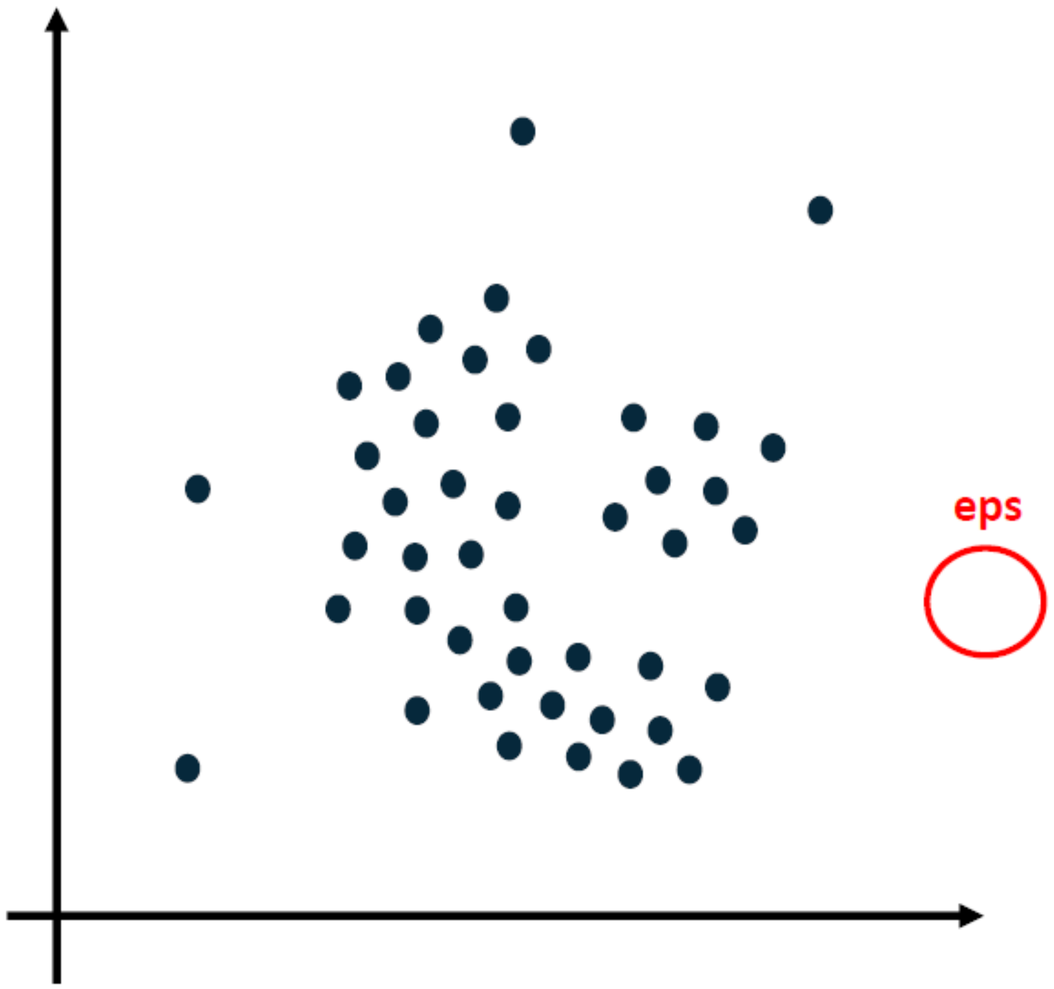
Here's the **DBSCAN algorithm**:

1. Label all data points as core, border, or noise points
2. Eliminate noise points
3. Connect all core points that are within eps distance of each other
4. Make each group of connected core points a separate cluster
5. Assign each border point to one of the clusters based on proximity to core points

1. Label all data points as core, border, or noise points

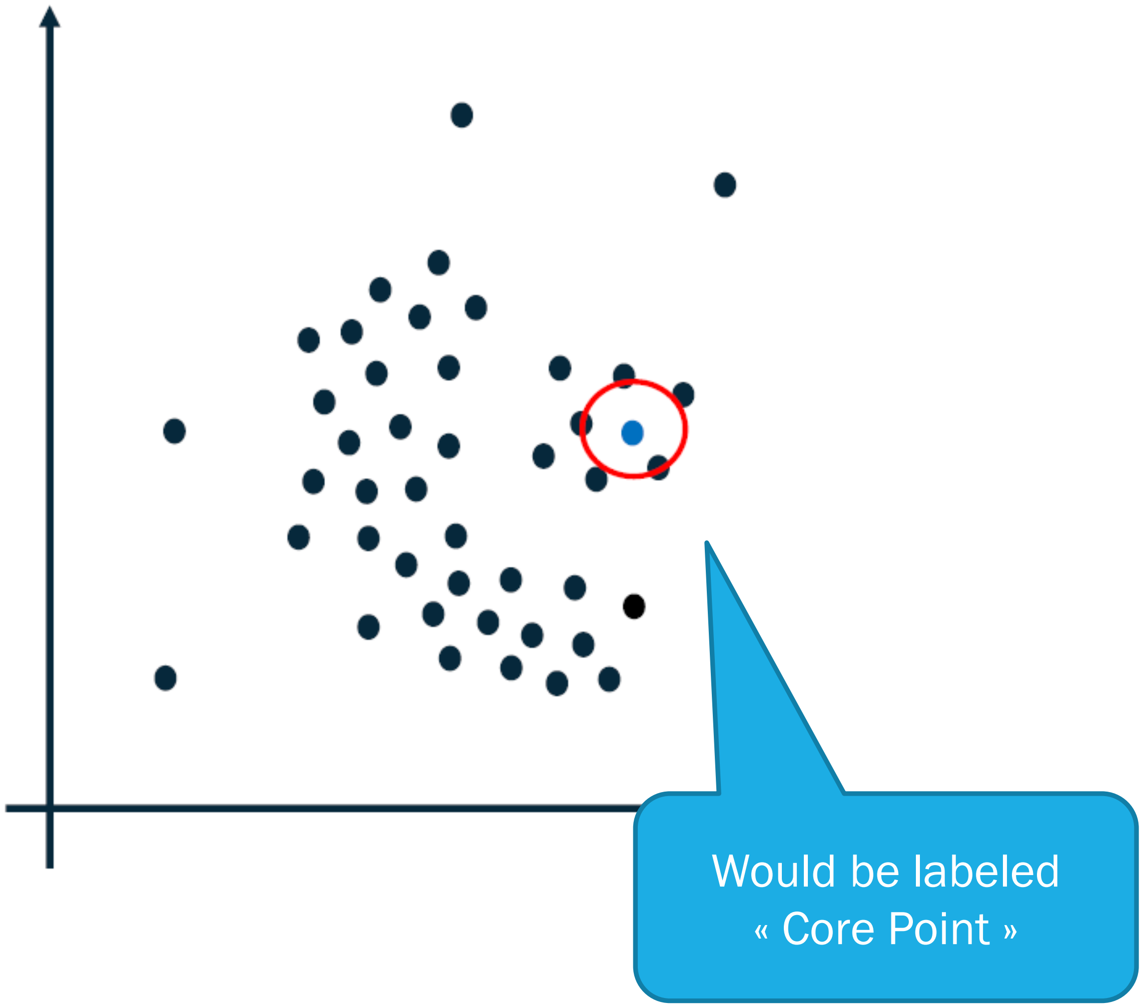


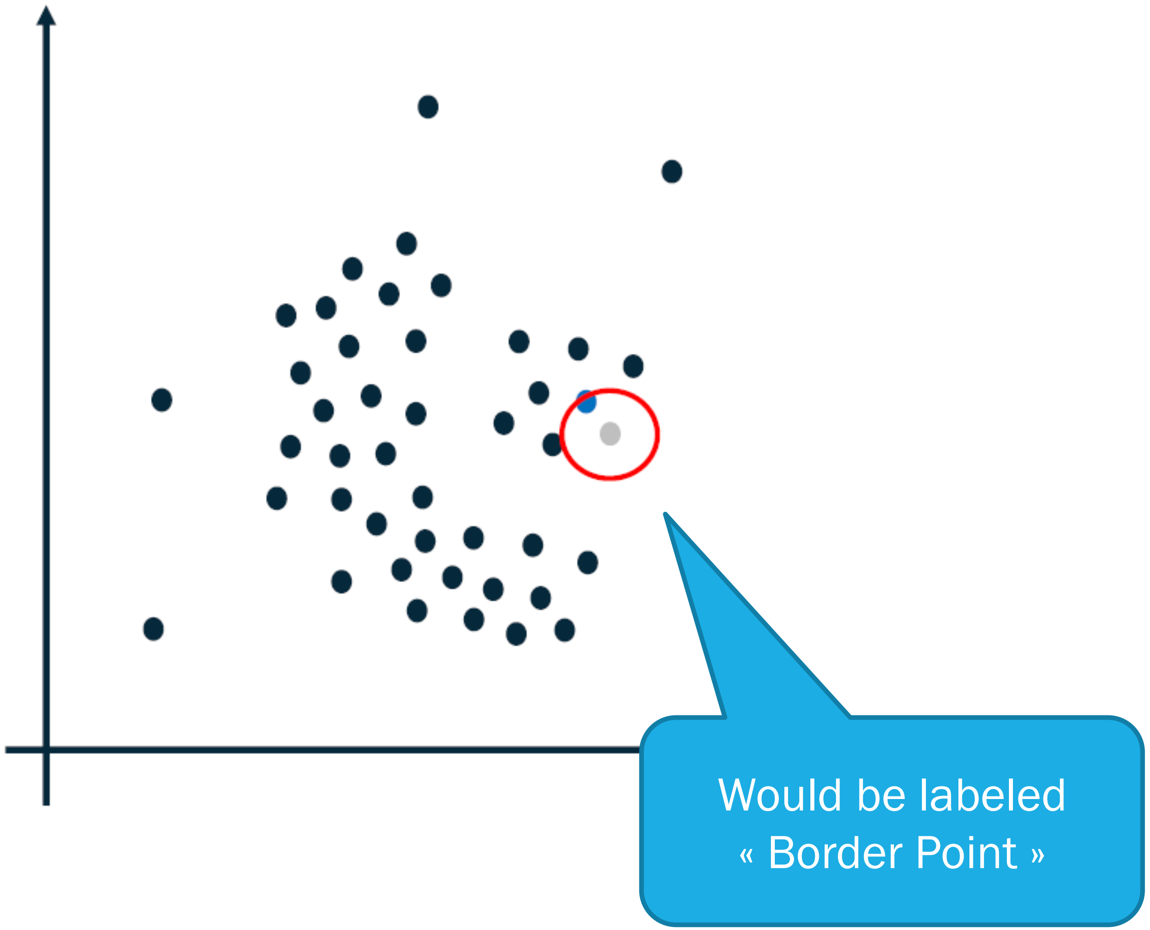
1. Label all data points as core, border, or noise points



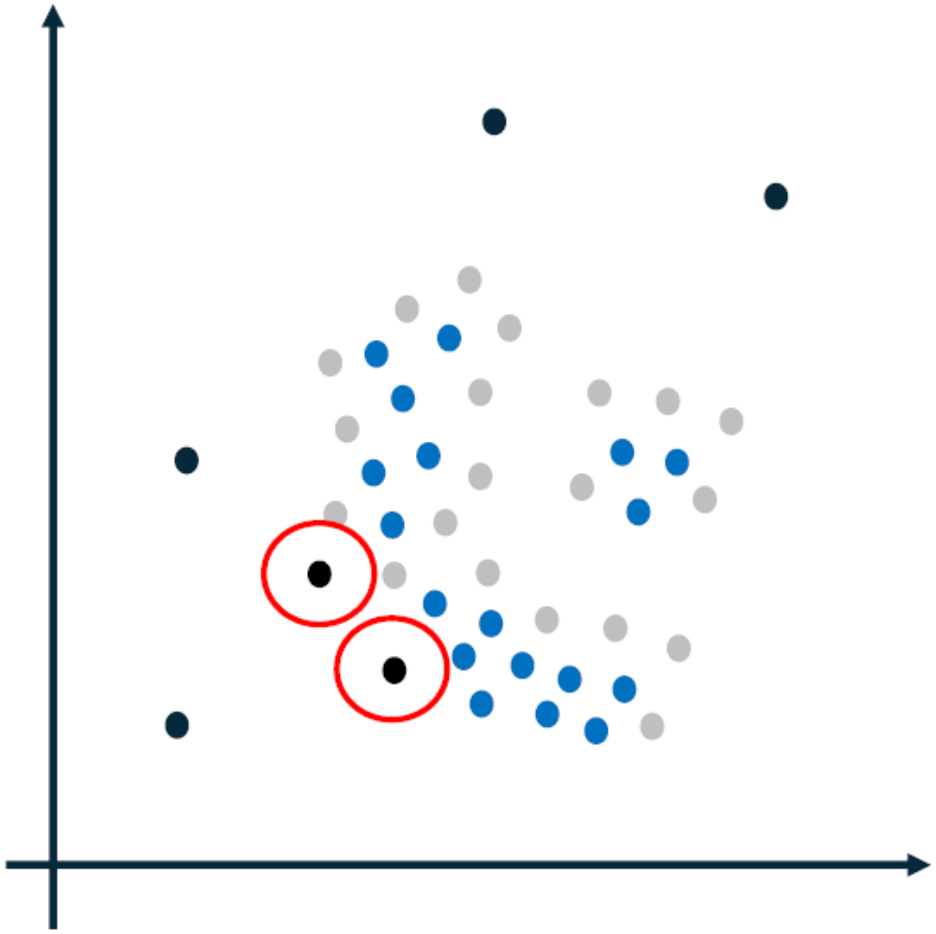
Decide on eps
Min_samples = 4
(includes the point itself)

1. Label all data points as core, border, or noise points





1. Label all data points as core, border, or noise points



Process continues for all points

Points that are not border nor core are « outlier points »

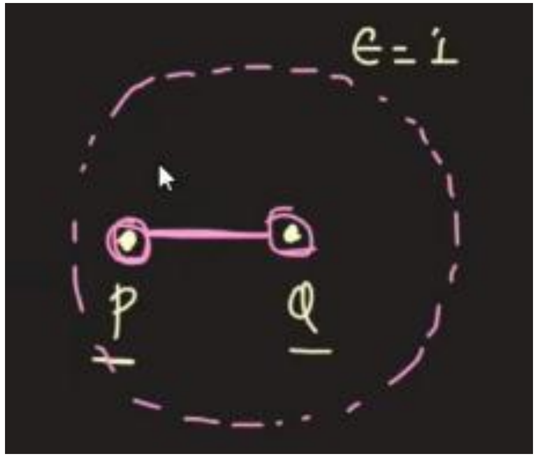
2. Eliminate noise points

3. Connect all core points that are within eps distance of each other

Directly Density Reachable :

A point P is directly density-reachable from a point Q given Eps, MinPts if:

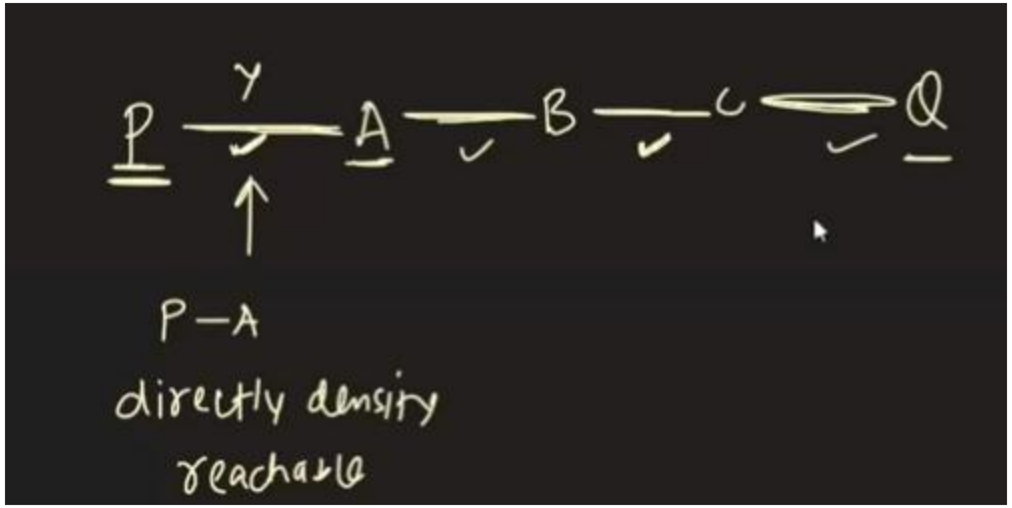
1. P is in the Eps-neighborhood of Q
2. Both P and Q are core points



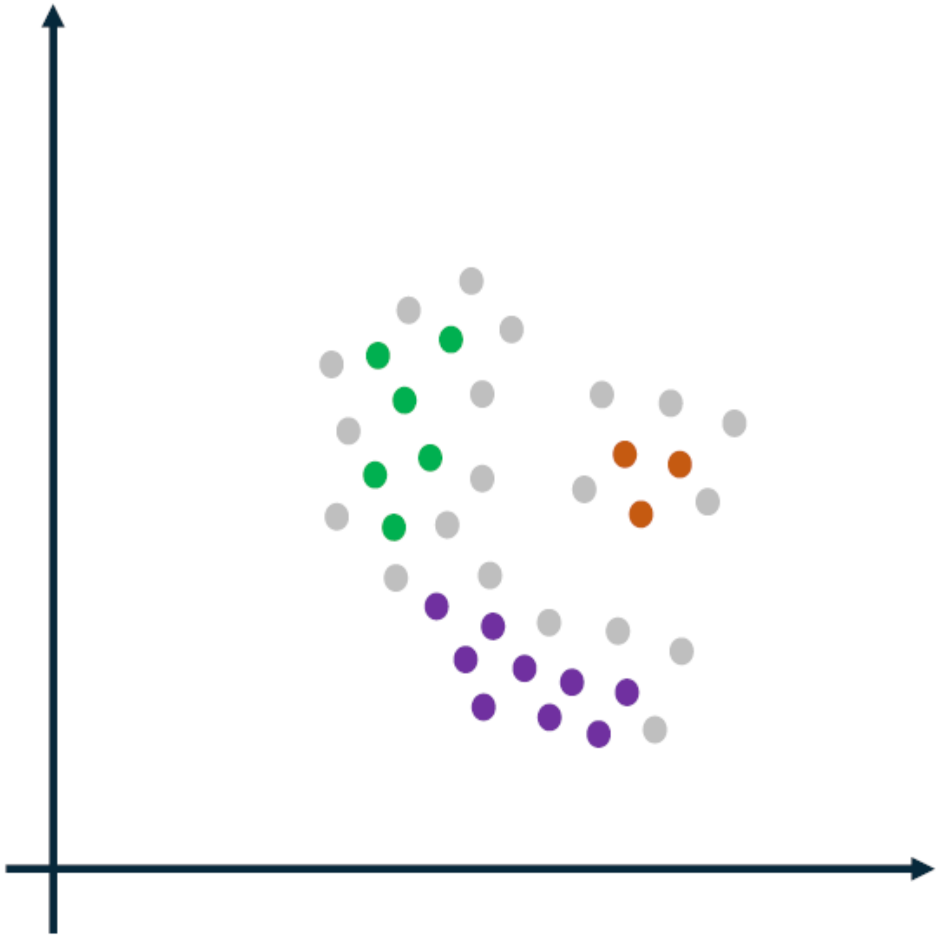
3. Connect all core points that are within eps distance of each other

Density Connected Points

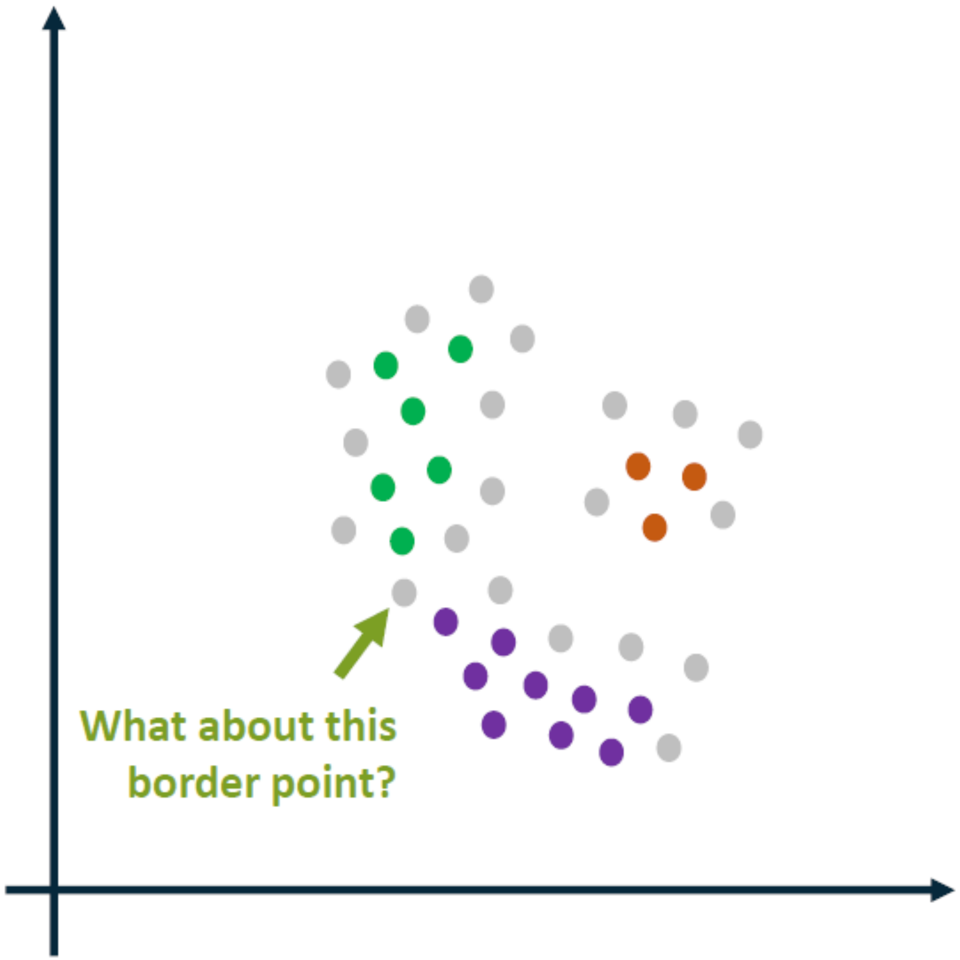
A point P is density connected to Q given Eps, MinPts if there is a chain of points P1, P2, P3 ...... Pn, P1 = P and Pn = Q such that Pi+1 is directly density reachable from Pi .



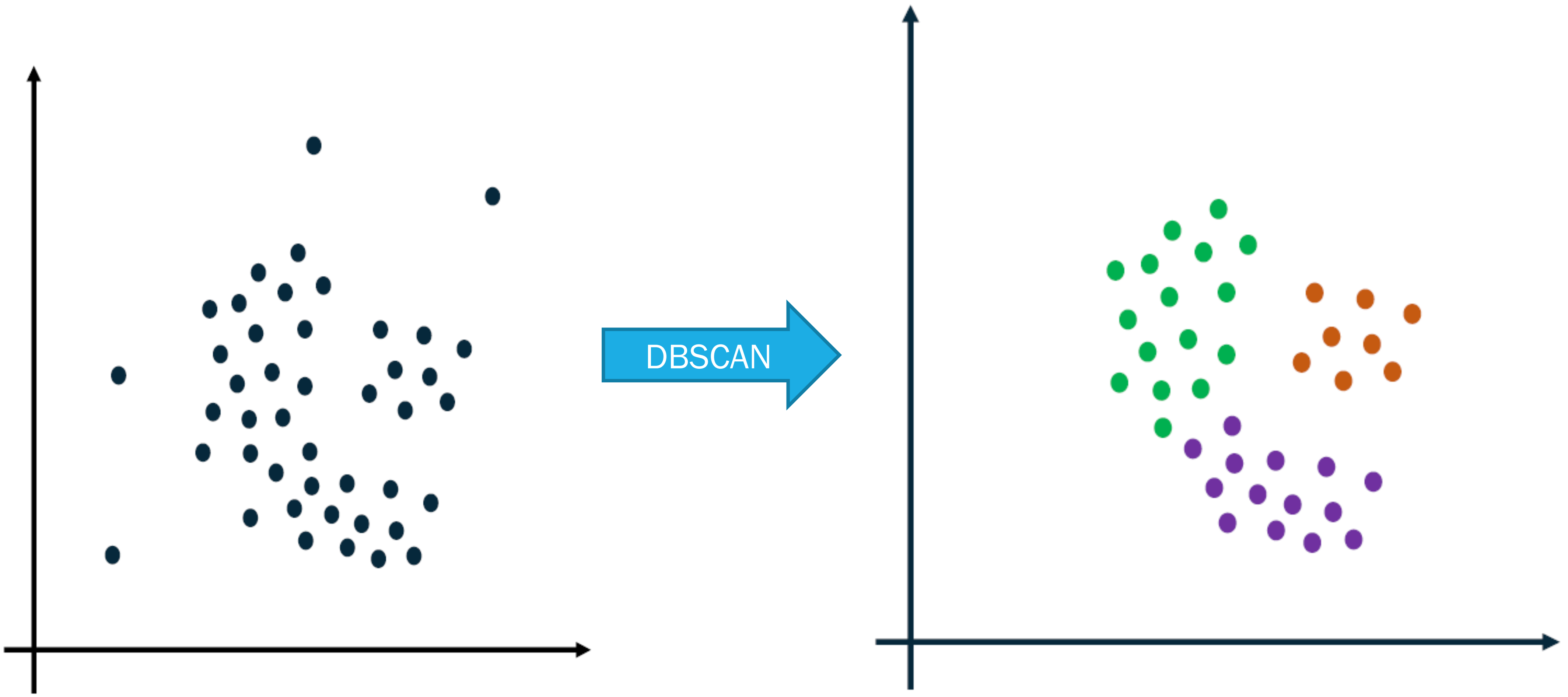
3. Connect all core points that are within eps distance of each other
4. Make each group of connected core points a separate cluster



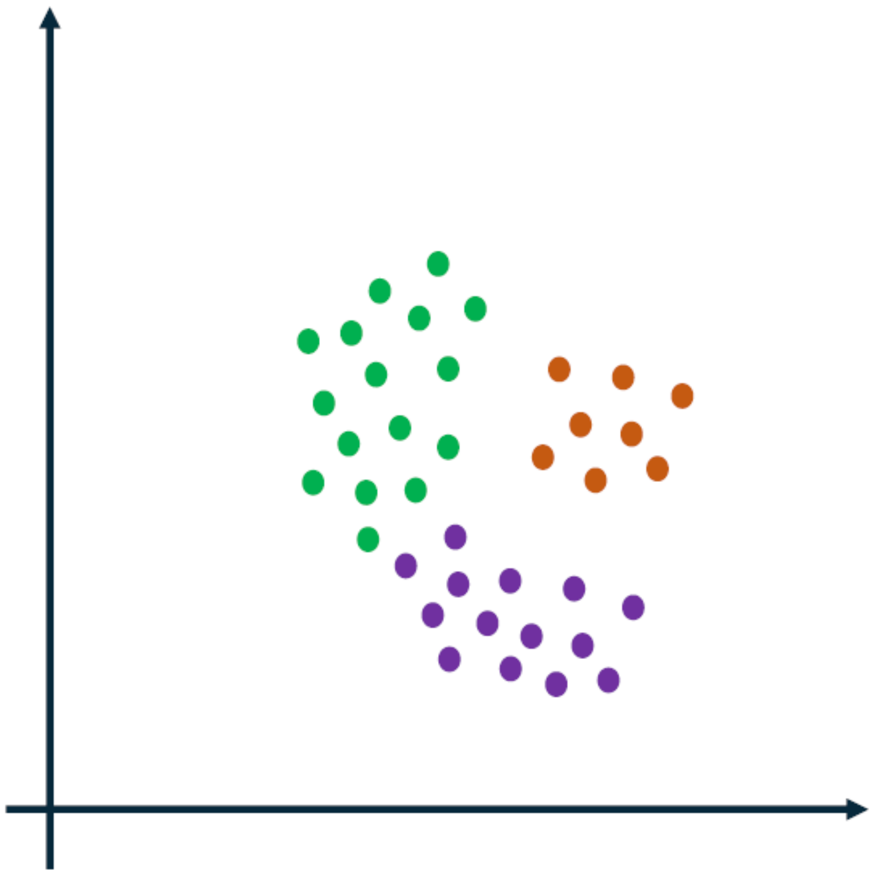
5. Assign each border point to one of the clusters based on proximity to core points

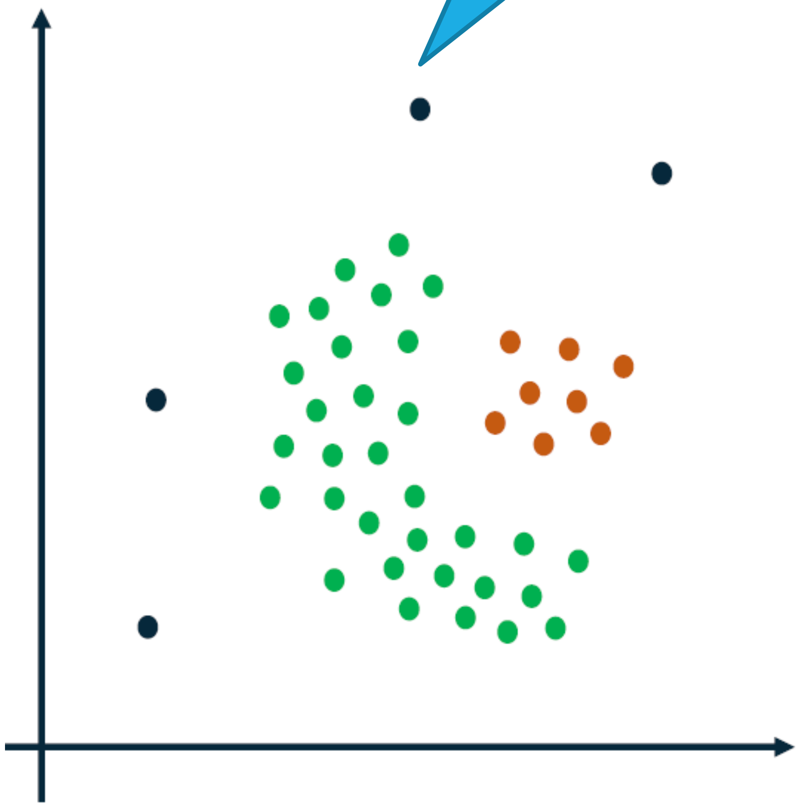


Greedy approach, if equal distance, assign border points to first created cluster


DBSCAN

Perhaps we would have expected only 2 from visual inspection.



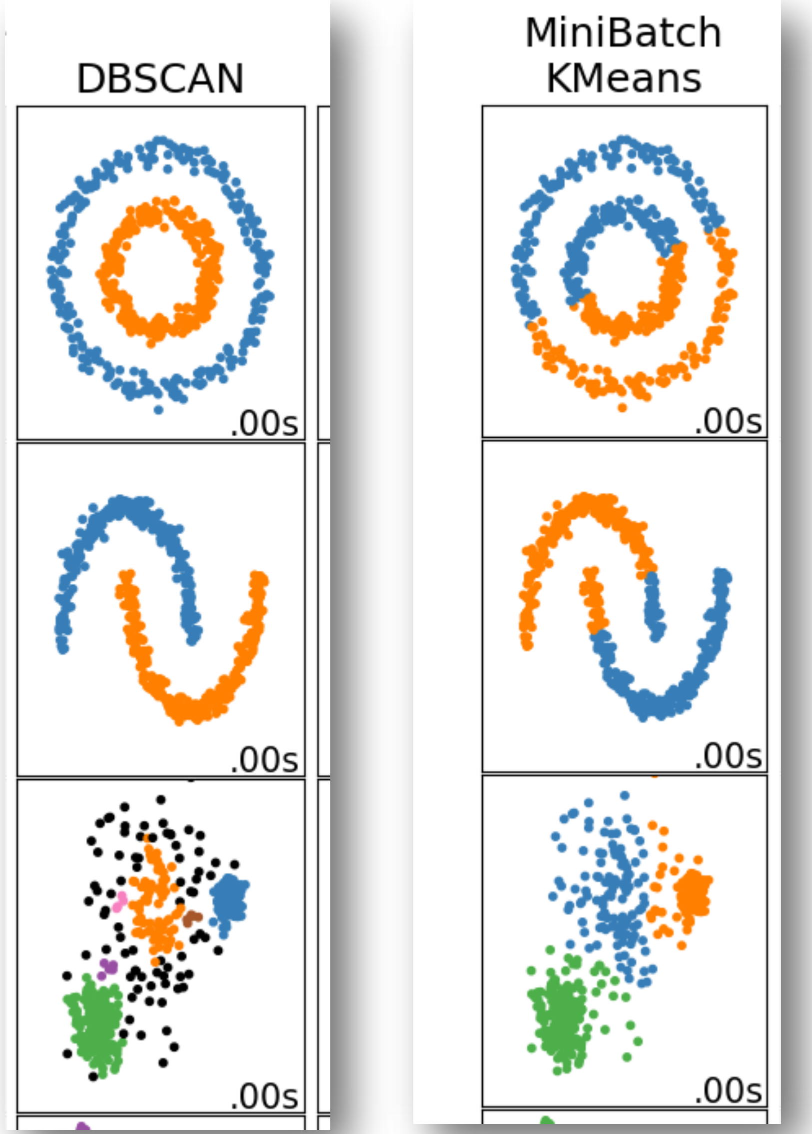


Evaluation

Source

Visual inspection

Check that the clusters look as you'd expect, and that examples you consider similar to each other appear in the same cluster.



These Kmeans metrics don’t necessarily apply

Cluster Metrics

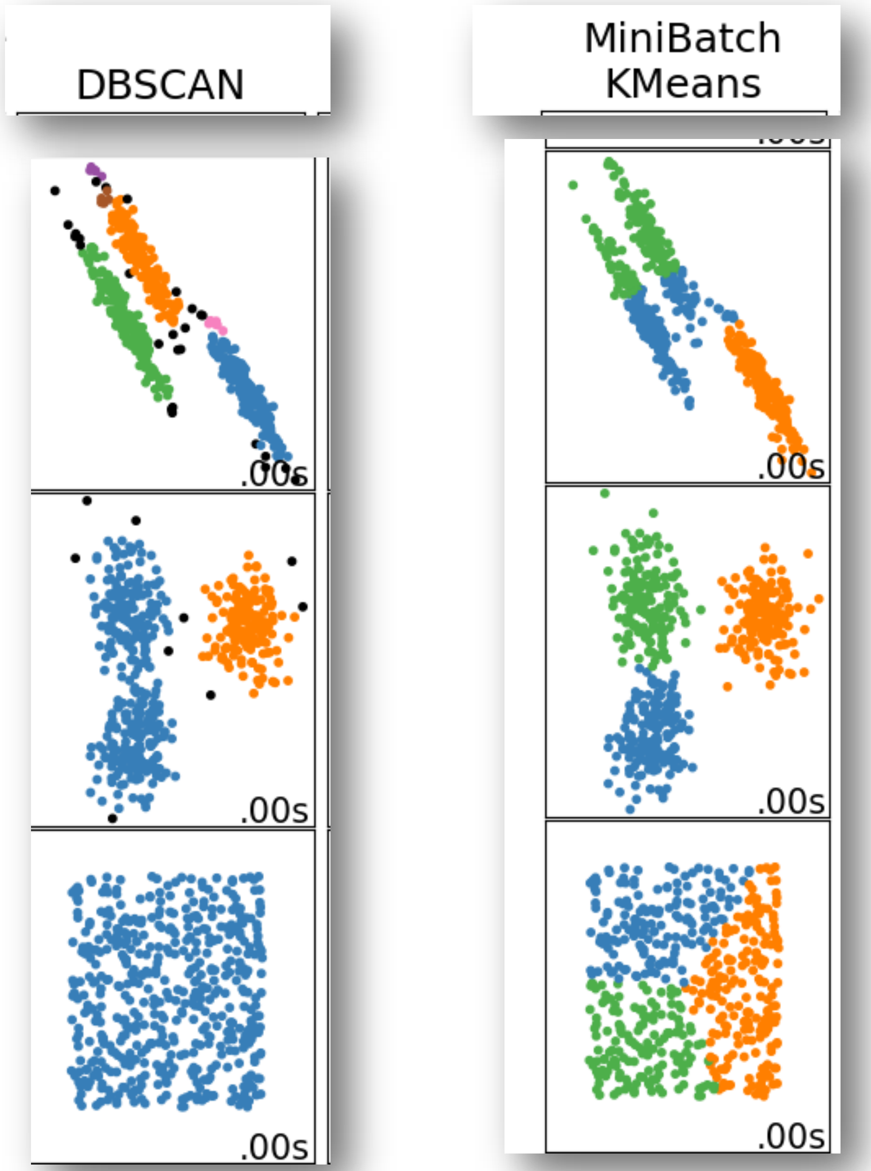
Possibly use these commonly-used metrics (not an exhaustive list):
- Cluster cardinality
- Cluster magnitude
- Overall Distance

Similarity Measure

If the clustering seems inadequate, perhaps reinvestigate the similarity measure.

Impact

Evaluate the downstream performance.



CLUSTERING

- Introduction
- Similarity measures
- Centroid-based clustering (KMeans)
- Density-based clustering (DBSCAN)